

Curve Sketching

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Coordinate Geometry (ref `cg5`)

This document contains all 70 *Curve Sketching* questions, each with a fully worked solution and a TikZ sketch of the curve, plotted from its equation. Roots, intercepts, turning points and points of inflection are marked, and asymptotes are drawn as dashed grey lines. (The few questions that only ask you to *describe* or *match* shapes, with several functions at once, have no single plot.)

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Curve Sketching 01 (cg5-001)

Difficulty: Foundation / Marks: 4

State the general shape of the graph of each of the following functions, giving a reason in each case.

- a) $y = 4x^5$
- b) $y = -2x^4$
- c) $y = -7x^3$
- d) $y = 3x^6$

Worked solution.

Step 1: For $y = 4x^5$: odd power, positive coefficient.

Power 5 is odd, $k = 4 > 0$

Odd power with positive k gives a bottom-left to top-right shape.

Step 2: For $y = -2x^4$: even power, negative coefficient.

Power 4 is even, $k = -2 < 0$

Even power with negative k gives an n-shape (opening downward).

Step 3: For $y = -7x^3$: odd power, negative coefficient.

Power 3 is odd, $k = -7 < 0$

Odd power with negative k gives a top-left to bottom-right shape.

Step 4: For $y = 3x^6$: even power, positive coefficient.

Power 6 is even, $k = 3 > 0$

Even power with positive k gives a u-shape (opening upward).

Final answer: a) Bottom-left to top-right; b) n-shape; c) Top-left to bottom-right; d) u-shape

Curve Sketching 02 (cg5-002)

Difficulty: Foundation / Marks: 4

Four graphs are shown below, labelled A, B, C, and D. Match each graph to the correct function.

- Graph A: u-shape entirely above the x -axis, passing through the origin. - Graph B: top-left to bottom-right, passing through the origin. - Graph C: n-shape entirely below the x -axis, passing through the origin. - Graph D: bottom-left to top-right, passing through the origin.

Functions: $y = 0.5x^3$, $y = -4x^2$, $y = 2x^4$, $y = -3x^5$

Worked solution.

Step 1: Graph A is a u-shape: even power, positive coefficient.

$$y = 2x^4 \quad (\text{even, positive})$$

x^4 is even so gives a u/n-shape; positive coefficient means u-shape.

Step 2: Graph B is top-left to bottom-right: odd power, negative coefficient.

$$y = -3x^5 \quad (\text{odd, negative})$$

Odd power with negative k gives top-left to bottom-right.

Step 3: Graph C is n-shape: even power, negative coefficient.

$$y = -4x^2 \quad (\text{even, negative})$$

Even power with negative k gives an n-shape below the axis.

Step 4: Graph D is bottom-left to top-right: odd power, positive coefficient.

$$y = 0.5x^3 \quad (\text{odd, positive})$$

Odd power with positive k gives bottom-left to top-right.

Final answer: $A \leftrightarrow y = 2x^4$; $B \leftrightarrow y = -3x^5$; $C \leftrightarrow y = -4x^2$; $D \leftrightarrow y = 0.5x^3$

Curve Sketching 03 (cg5-003)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = x(x - 2)(x + 3)$, clearly showing where the curve meets the coordinate axes.

Worked solution.

Step 1: Identify the leading term by expanding partially.

$$y = x \cdot x \cdot x + \dots = x^3 + \dots$$

The highest power is x^3 with a positive coefficient, so the shape is bottom-left to top-right.

Step 2: Find the y -intercept by substituting $x = 0$.

$$y = 0(0 - 2)(0 + 3) = 0$$

The curve passes through the origin.

Step 3: Find the x -intercepts by setting each factor to zero.

$$x = 0, \quad x - 2 = 0 \Rightarrow x = 2, \quad x + 3 = 0 \Rightarrow x = -3$$

Three distinct roots at $x = -3, 0, 2$.

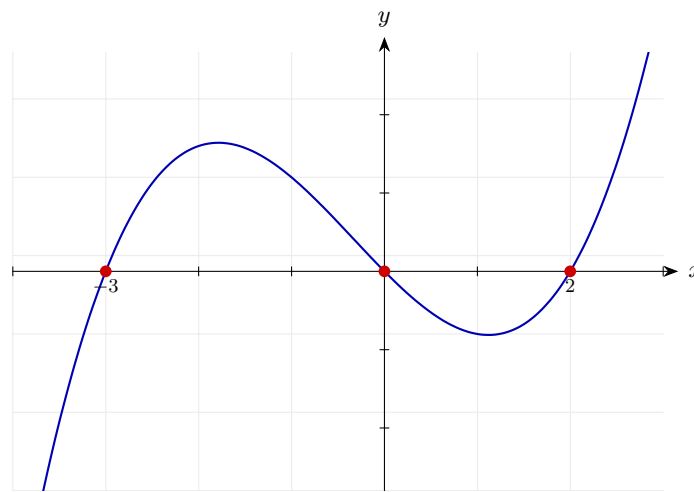
Step 4: Describe the sketch.

Crosses x -axis at $-3, 0, 2$; y -intercept at 0

The curve crosses all three roots (no repeated factors), rising from bottom-left to top-right overall.

Final answer: Cubic crossing x -axis at $x = -3, 0, 2$; passes through origin; positive leading coefficient.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 04 (cg5-004)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = (x + 1)(x - 3)(x - 5)$, showing clearly where the curve meets the axes.

Worked solution.

Step 1: Determine the overall shape.

$$y = x^3 + \dots \quad (\text{positive leading coefficient})$$

Bottom-left to top-right shape.

Step 2: Find the y -intercept.

$$y = (1)(-3)(-5) = 15$$

Substitute $x = 0$: the three factors give $1 \times (-3) \times (-5) = 15$.

Step 3: Find the x -intercepts.

$$x + 1 = 0 \Rightarrow x = -1; \quad x - 3 = 0 \Rightarrow x = 3; \quad x - 5 = 0 \Rightarrow x = 5$$

Three distinct roots at $x = -1, 3, 5$.

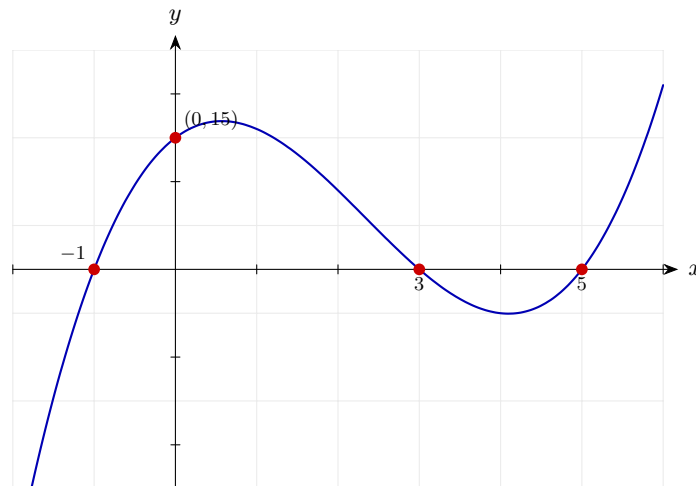
Step 4: Describe the sketch.

Crosses at $x = -1, 3, 5$; y -intercept at $(0, 15)$

The curve crosses each root cleanly, with y -intercept at 15.

Final answer: Cubic crossing x -axis at $-1, 3, 5$; y -intercept at $(0, 15)$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 05 (cg5-005)

Difficulty: Foundation | Marks: 4

Sketch the graph of $y = -x(x + 2)(x - 4)$, showing where the curve meets the axes.

Worked solution.

Step 1: Identify the leading term.

$$y = -x \cdot x \cdot x + \dots = -x^3 + \dots$$

Negative coefficient of x^3 means top-left to bottom-right shape.

Step 2: Find the y -intercept.

$$y = -(0)(2)(-4) = 0$$

The curve passes through the origin.

Step 3: Find the x -intercepts.

$$x = 0, \quad x = -2, \quad x = 4$$

Each factor set to zero gives a distinct root.

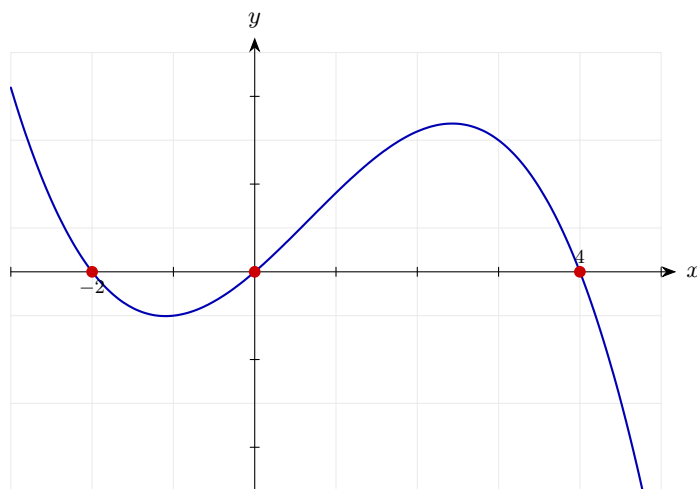
Step 4: Describe the sketch.

Crosses at $x = -2, 0, 4$; top-left to bottom-right overall

Negative leading coefficient flips the orientation compared to a standard cubic.

Final answer: Cubic crossing x -axis at $x = -2, 0, 4$; passes through origin; top-left to bottom-right shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 06 (cg5-006)

Difficulty: Foundation / Marks: 5

a) Factorise completely $x^3 + x^2 - 6x$.

b) Hence sketch the graph of $y = x^3 + x^2 - 6x$, **showing where it meets the coordinate axes.**

Worked solution.

Step 1: Part a: take out the common factor x .

$$x^3 + x^2 - 6x = x(x^2 + x - 6)$$

Every term has a factor of x .

Step 2: Factorise the quadratic $x^2 + x - 6$.

$$x^2 + x - 6 = (x + 3)(x - 2)$$

We need two numbers that multiply to -6 and add to 1 : these are 3 and -2 .

Step 3: Write the full factorisation.

$$x(x + 3)(x - 2)$$

Three linear factors, so three roots.

Step 4: Part b: find x -intercepts and y -intercept.

$$x = 0, -3, 2; \quad y(0) = 0$$

All three roots are distinct; the y -intercept is at the origin.

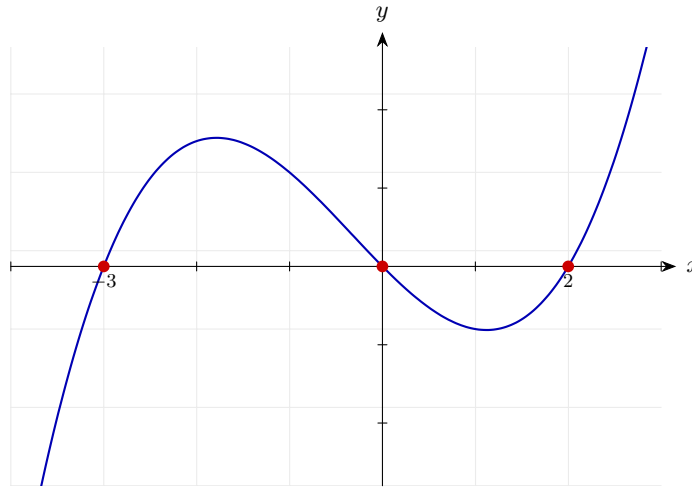
Step 5: Describe the sketch.

Positive x^3 coefficient: bottom-left to top-right

The curve crosses all three roots and rises to the right.

Final answer: a) $x(x+3)(x-2)$; b) Cubic crossing at $x = -3, 0, 2$; positive shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 07 (cg5-007)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = (2x+1)(x-2)(x-4)$, showing where the curve meets the axes.

Worked solution.

Step 1: Find the leading coefficient.

$$2x \cdot x \cdot x = 2x^3 \quad (\text{positive})$$

Positive leading coefficient: bottom-left to top-right shape.

Step 2: Find the y-intercept.

$$y = (1)(-2)(-4) = 8$$

Substitute $x = 0$ into each factor.

Step 3: Find the x-intercepts.

$$2x + 1 = 0 \Rightarrow x = -\frac{1}{2}; \quad x = 2; \quad x = 4$$

Three distinct roots including a fractional one.

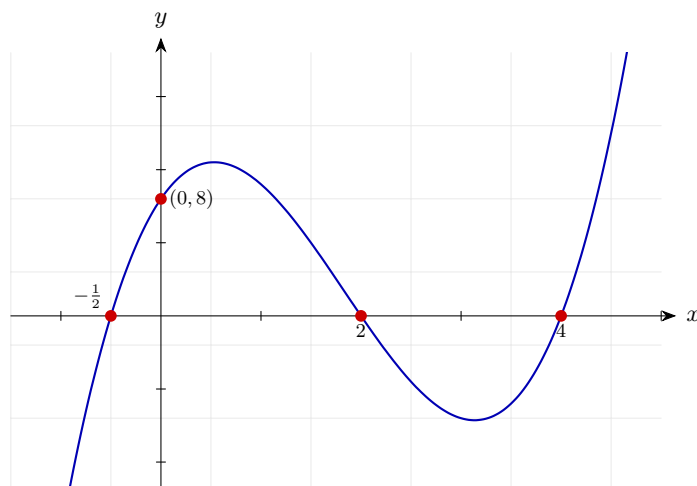
Step 4: Describe the sketch.

$$\text{Crosses at } x = -\frac{1}{2}, 2, 4; \quad y\text{-intercept at } (0, 8)$$

The curve crosses all three roots cleanly.

Final answer: Cubic crossing x -axis at $x = -\frac{1}{2}, 2, 4$; y -intercept at $(0, 8)$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 08 (cg5-008)

Difficulty: Foundation / Marks: 5

a) Factorise completely $y = 2x^3 - 8x$.

b) Hence sketch the graph, showing where it meets the coordinate axes.

Worked solution.

Step 1: Part a: take out the common factor.

$$2x^3 - 8x = 2x(x^2 - 4)$$

Factor out $2x$.

Step 2: Factorise $x^2 - 4$ as a difference of two squares.

$$x^2 - 4 = (x - 2)(x + 2)$$

$a^2 - b^2 = (a - b)(a + b)$ with $a = x, b = 2$.

Step 3: Write the full factorisation.

$$2x(x - 2)(x + 2)$$

Three distinct linear factors.

Step 4: Part b: roots and y-intercept.

$$x = 0, 2, -2; \quad y(0) = 0$$

Symmetrically placed roots at ± 2 and at the origin.

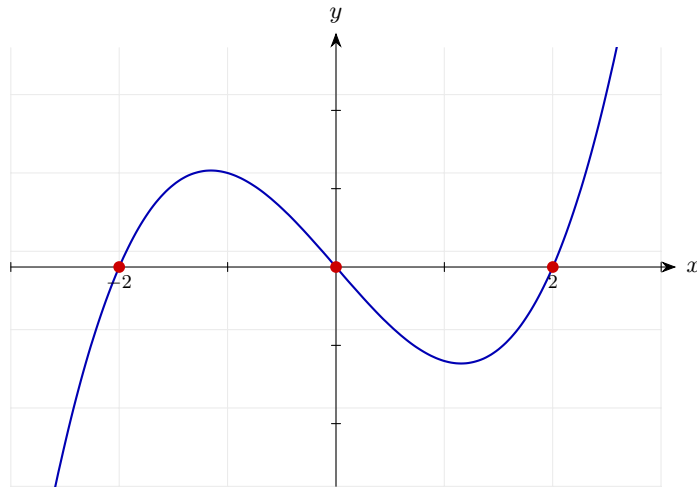
Step 5: Describe the sketch.

Positive leading coefficient: bottom-left to top-right

The graph is symmetric about the origin (odd function) and crosses at $-2, 0, 2$.

Final answer: a) $2x(x - 2)(x + 2)$; b) Cubic crossing at $x = -2, 0, 2$; passes through origin.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 09 (cg5-009)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = x^2(x - 3)$, showing clearly where the curve meets the axes and describing the behaviour at each x -intercept.

Worked solution.

Step 1: Identify the shape from the leading term.

$$y = x^3 - 3x^2 \implies \text{positive } x^3$$

Bottom-left to top-right overall shape.

Step 2: Find the y -intercept.

$$y = 0^2(0 - 3) = 0$$

Passes through the origin.

Step 3: Find the x -intercepts and their nature.

$$x^2 = 0 \Rightarrow x = 0 \text{ (double root); } x - 3 = 0 \Rightarrow x = 3 \text{ (single root)}$$

A double root means the curve touches the x -axis at $x = 0$ without crossing it.

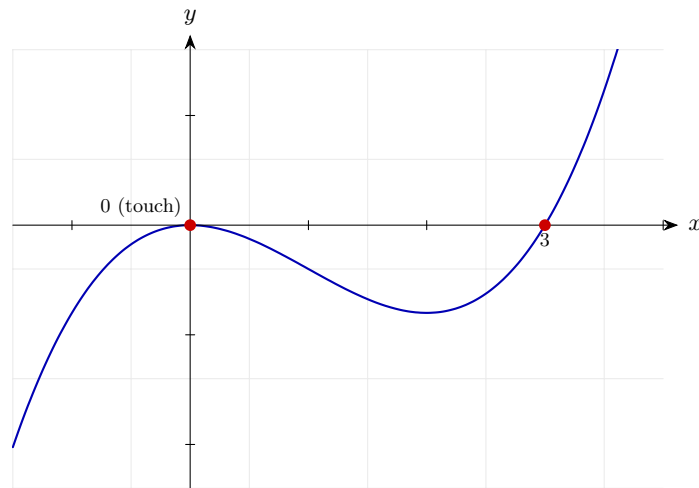
Step 4: Describe the behaviour at each root.

$$x = 0 : \text{ touches; } x = 3 : \text{ crosses}$$

The curve bounces off the axis at $x = 0$ and cuts through at $x = 3$.

Final answer: Cubic touching x -axis at $x = 0$ (double root) and crossing at $x = 3$; y -intercept at origin.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 10 (cg5-010)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = (x + 2)^2(x - 1)$, showing where the curve meets the axes.

Worked solution.

Step 1: Identify the leading term.

$$x^2 \cdot x = x^3 \quad (\text{positive coefficient})$$

Bottom-left to top-right shape.

Step 2: Find the y -intercept.

$$y = (0 + 2)^2(0 - 1) = 4 \times (-1) = -4$$

The curve crosses the y -axis at -4 .

Step 3: Find the x -intercepts.

$$(x + 2)^2 = 0 \Rightarrow x = -2 \text{ (double); } \quad x - 1 = 0 \Rightarrow x = 1 \text{ (single)}$$

The curve touches at $x = -2$ and crosses at $x = 1$.

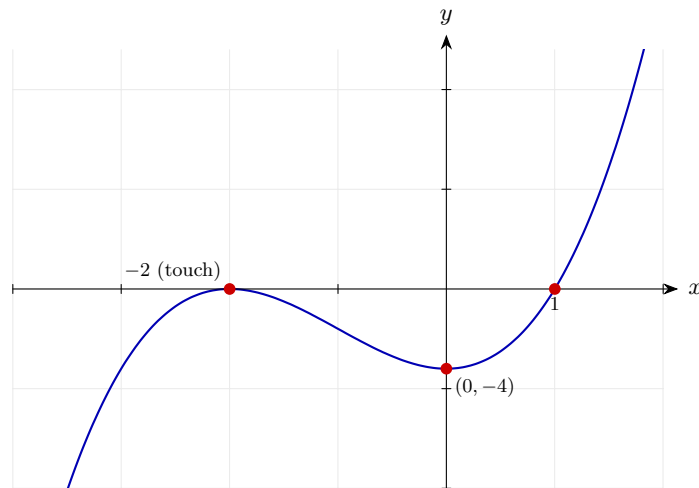
Step 4: Describe the sketch.

Touches at $x = -2$, crosses at $x = 1$, y -intercept at $(0, -4)$

The curve dips down to touch the axis at -2 and rises to cross at $x = 1$.

Final answer: Cubic touching x -axis at $x = -2$, crossing at $x = 1$; y -intercept at $(0, -4)$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 11 (cg5-011)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = (3 - x)^3$, showing where the curve meets the axes and describing the behaviour at the x -intercept.

Worked solution.

Step 1: Expand to identify the leading term.

$$(3 - x)^3 = 27 - 27x + 9x^2 - x^3$$

The leading term is $-x^3$: negative coefficient, so top-left to bottom-right shape.

Step 2: Find the y -intercept.

$$y = (3 - 0)^3 = 27$$

The curve crosses the y -axis at 27.

Step 3: Find the x -intercept.

$$(3 - x)^3 = 0 \Rightarrow x = 3 \text{ (triple root)}$$

A triple root still crosses the x -axis, but the curve flattens out as it crosses.

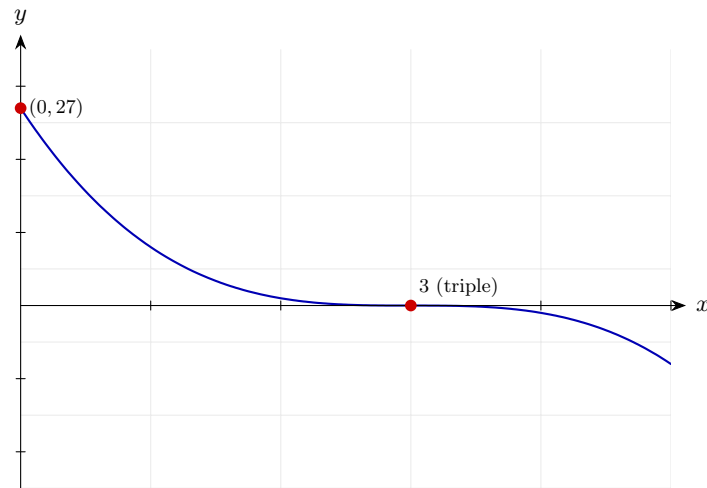
Step 4: Describe the sketch.

Enters top-left, passes through $(0, 27)$, flattens and crosses at $x = 3$

The triple root gives an inflection-type crossing, not a sharp cut.

Final answer: Cubic with triple root at $x = 3$; y -intercept at $(0, 27)$; top-left to bottom-right overall shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 12 (cg5-012)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = -x^2(x+4)$, showing where the curve meets the axes and describing the behaviour at each x -intercept.

Worked solution.

Step 1: Identify the leading term.

$$-x^2 \cdot x = -x^3 \quad (\text{negative})$$

Top-left to bottom-right overall shape.

Step 2: Find the y -intercept.

$$y = -(0)^2(0+4) = 0$$

The curve passes through the origin.

Step 3: Find the x -intercepts.

$$x^2 = 0 \Rightarrow x = 0 \text{ (double); } x + 4 = 0 \Rightarrow x = -4 \text{ (single)}$$

Touches at $x = 0$ and crosses at $x = -4$.

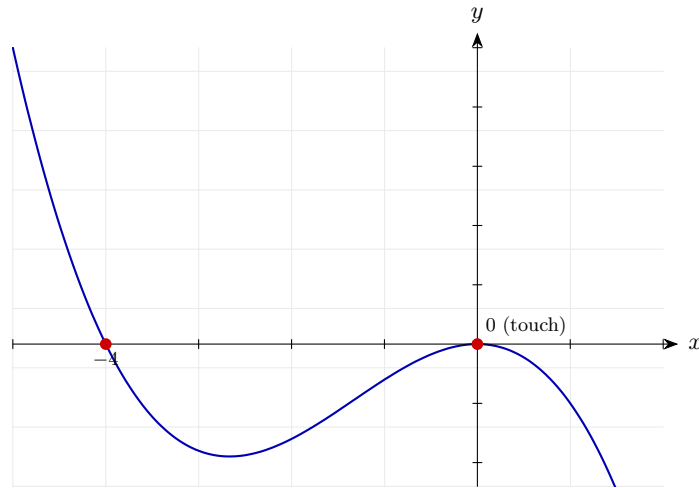
Step 4: Describe the sketch.

Crosses at $x = -4$, touches at $x = 0$; falls to bottom-right

The curve enters from top-left, crosses at -4 , rises to touch the origin, then falls away to the right.

Final answer: Cubic crossing at $x = -4$, touching at $x = 0$; passes through origin; top-left to bottom-right shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 13 (cg5-013)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = x(x - 5)^2$, showing all intercepts and describing the nature of each root.

Worked solution.

Step 1: Identify the leading term.

$$x \cdot x^2 = x^3 \quad (\text{positive})$$

Bottom-left to top-right overall.

Step 2: Find the y-intercept.

$$y = 0 \cdot (0 - 5)^2 = 0$$

Passes through the origin.

Step 3: Find the x-intercepts.

$$x = 0 \text{ (single root); } (x - 5)^2 = 0 \Rightarrow x = 5 \text{ (double root)}$$

Crosses at $x = 0$, touches at $x = 5$.

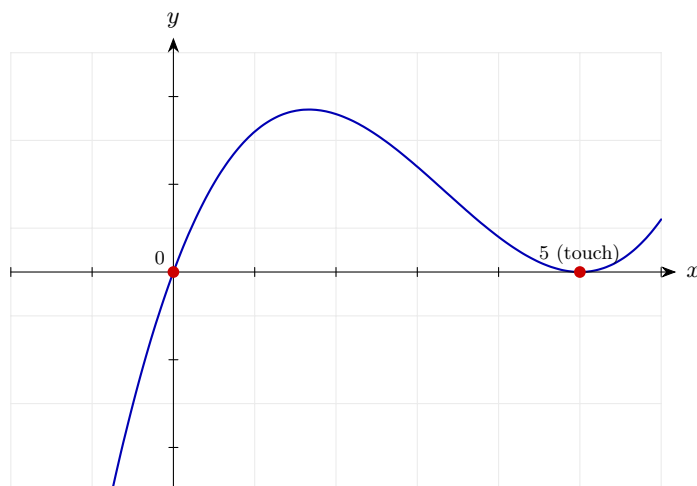
Step 4: Describe the sketch.

Crosses at 0, rises, touches at 5 (bouncing off), then rises away

The curve crosses the origin, climbs to a local maximum, dips to touch $x = 5$ and rises again.

Final answer: Cubic crossing at $x = 0$, touching at $x = 5$; y -intercept at origin; positive shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 14 (cg5-014)

Difficulty: Foundation / Marks: 5

a) Factorise completely $2x^3 - 12x^2 + 18x$.

b) Hence sketch the graph, showing where it meets the coordinate axes.

Worked solution.

Step 1: Part a: take out the common factor.

$$2x^3 - 12x^2 + 18x = 2x(x^2 - 6x + 9)$$

Each term has a factor of $2x$.

Step 2: Factorise the quadratic.

$$x^2 - 6x + 9 = (x - 3)^2$$

Perfect square: $(x - 3)^2 = x^2 - 6x + 9$.

Step 3: Write the full factorisation.

$$2x(x - 3)^2$$

One single root at $x = 0$, one double root at $x = 3$.

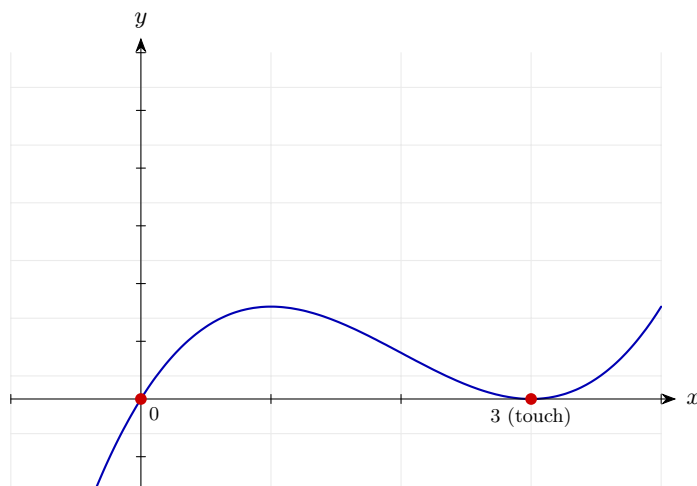
Step 4: Part b: find intercepts.

$$x = 0 \text{ (crosses), } x = 3 \text{ (touches); } y(0) = 0$$

Positive leading coefficient: bottom-left to top-right.

Final answer: a) $2x(x - 3)^2$; b) Crosses at $x = 0$, touches at $x = 3$; positive cubic shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 15 (cg5-015)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = x(x - 1)(x + 1)(x - 3)$, showing where the curve meets the axes.

Worked solution.

Step 1: Identify the degree and leading coefficient.

$$x \cdot x \cdot x \cdot x = x^4 \quad (\text{positive})$$

Even degree, positive leading coefficient: u-shape overall.

Step 2: Find the y-intercept.

$$y = 0(-1)(1)(-3) = 0$$

The curve passes through the origin.

Step 3: Find the x-intercepts.

$$x = 0, -1, 1, 3$$

Four distinct real roots — the curve crosses the axis four times.

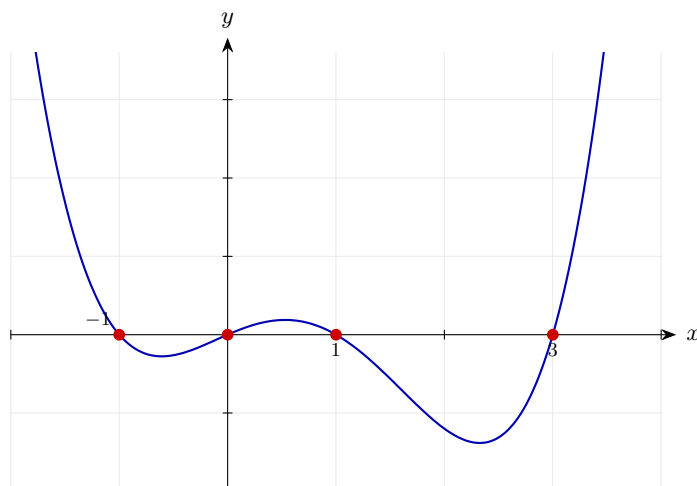
Step 4: Describe the sketch.

Quartic crossing at $x = -1, 0, 1, 3$; u-shape overall

The curve rises from bottom-left, crosses and undulates through all four roots, and rises again to the right.

Final answer: Quartic crossing at $x = -1, 0, 1, 3$; y-intercept at origin; u-shape overall.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 16 (cg5-016)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = -2x(x + 3)(x - 1)(x - 4)$, showing where the curve meets the axes.

Worked solution.

Step 1: Identify the leading term.

$$-2x \cdot x \cdot x \cdot x = -2x^4 \quad (\text{negative})$$

Even degree, negative leading coefficient: n-shape overall (opens downward).

Step 2: Find the y-intercept.

$$y = -2(0)(3)(-1)(-4) = 0$$

Passes through the origin.

Step 3: Find the x-intercepts.

$$x = 0, -3, 1, 4$$

Four distinct roots.

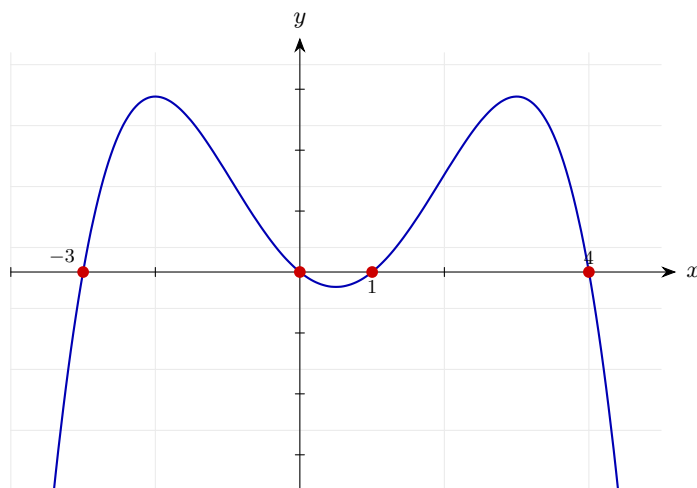
Step 4: Describe the sketch.

n-shape crossing at $x = -3, 0, 1, 4$

The curve falls from top-left, crossing and undulating through the four roots, then falls to the right.

Final answer: Quartic crossing at $x = -3, 0, 1, 4$; y -intercept at origin; n-shape overall.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 17 (cg5-017)

Difficulty: Foundation / Marks: 5

Sketch the graph of $y = x^2(x + 2)(x - 3)$, showing all intercepts and the nature of each root.

Worked solution.

Step 1: Identify the leading term.

$$x^2 \cdot x \cdot x = x^4 \quad (\text{positive})$$

u-shape overall.

Step 2: Find the y-intercept.

$$y = 0^2(2)(-3) = 0$$

Passes through the origin.

Step 3: Find the x-intercepts and their nature.

$$x = 0 \text{ (double, touches); } x = -2 \text{ (crosses); } x = 3 \text{ (crosses)}$$

The double root at $x = 0$ means the curve touches the axis there.

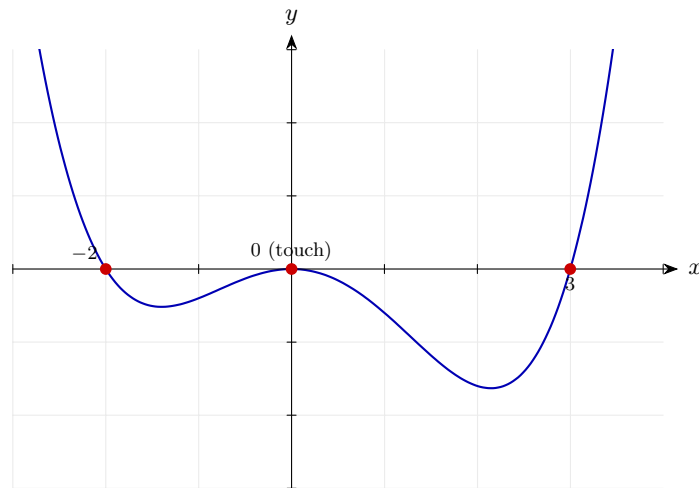
Step 4: Describe the sketch.

Crosses at -2 and 3 , touches at 0

The curve enters from bottom-left, crosses at -2 , touches origin, then crosses at 3 and rises.

Final answer: Quartic touching at $x = 0$, crossing at $x = -2$ and $x = 3$; y -intercept at origin.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 18 (cg5-018)

Difficulty: Foundation / Marks: 5

Sketch the graph of $y = (x-2)^2(x+1)^2$, showing all intercepts and describing the behaviour at each.

Worked solution.

Step 1: Identify the leading term.

$$x^2 \cdot x^2 = x^4 \quad (\text{positive})$$

u-shape overall.

Step 2: Find the y-intercept.

$$y = (0-2)^2(0+1)^2 = 4 \times 1 = 4$$

Substitute $x = 0$.

Step 3: Find the x-intercepts.

$$(x-2)^2 = 0 \Rightarrow x = 2 \text{ (double)}; \quad (x+1)^2 = 0 \Rightarrow x = -1 \text{ (double)}$$

Both roots are double roots — the curve touches but does not cross the x -axis at both.

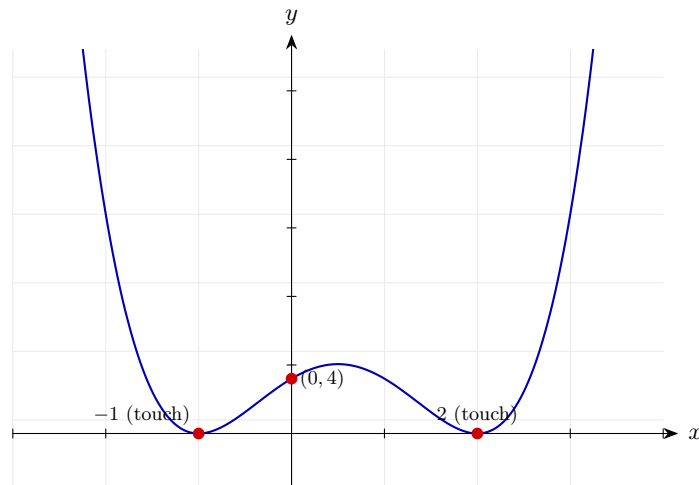
Step 4: Describe the sketch.

Touches at $x = -1$ and $x = 2$; y -intercept at $(0, 4)$

The entire curve sits on or above the x -axis (since $y \geq 0$ for all x).

Final answer: Quartic touching x -axis at $x = -1$ and $x = 2$; y -intercept at $(0, 4)$; curve lies above x -axis.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 19 (cg5-019)

Difficulty: Foundation / Marks: 6

a) Expand $(x^2 - 9)(x^2 - 4)$.

b) Hence sketch the graph of $y = (x^2 - 9)(x^2 - 4)$, **showing where it meets the axes.**

Worked solution.

Step 1: Part a: factorise each factor as a difference of two squares.

$$(x^2 - 9)(x^2 - 4) = (x - 3)(x + 3)(x - 2)(x + 2)$$

$$x^2 - 9 = (x - 3)(x + 3) \text{ and } x^2 - 4 = (x - 2)(x + 2).$$

Step 2: Expand fully (optional, for identification).

$$x^4 - 13x^2 + 36$$

Leading term is x^4 (positive): u-shape overall.

Step 3: Part b: find the x-intercepts.

$$x = \pm 3, \pm 2$$

Four distinct roots, symmetric about the y-axis.

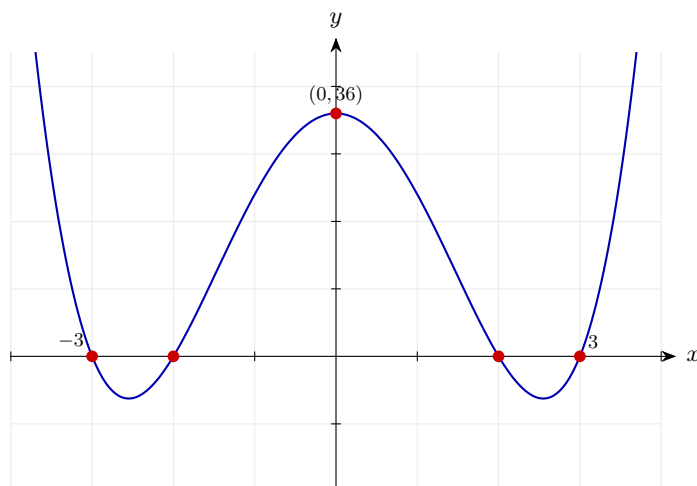
Step 4: Find the y-intercept.

$$y = (-9)(-4) = 36$$

Substitute $x = 0$ into the original factored form.

Final answer: Quartic crossing at $x = \pm 2, \pm 3$; y-intercept at $(0, 36)$; u-shape overall; symmetric about the y-axis.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 20 (cg5-020)

Difficulty: Foundation / Marks: 5

Sketch the graph of $y = 3x(x - 2)^2(x + 1)$, showing all intercepts and the nature of each.

Worked solution.

Step 1: Identify the leading term.

$$3x \cdot x^2 \cdot x = 3x^4 \quad (\text{positive})$$

u-shape overall.

Step 2: Find the y-intercept.

$$y = 3(0)(0 - 2)^2(0 + 1) = 0$$

Passes through the origin.

Step 3: Find the x-intercepts.

$$x = 0 \text{ (crosses); } x = 2 \text{ (double, touches); } x = -1 \text{ (crosses)}$$

Three distinct x -values; $x = 2$ is a double root.

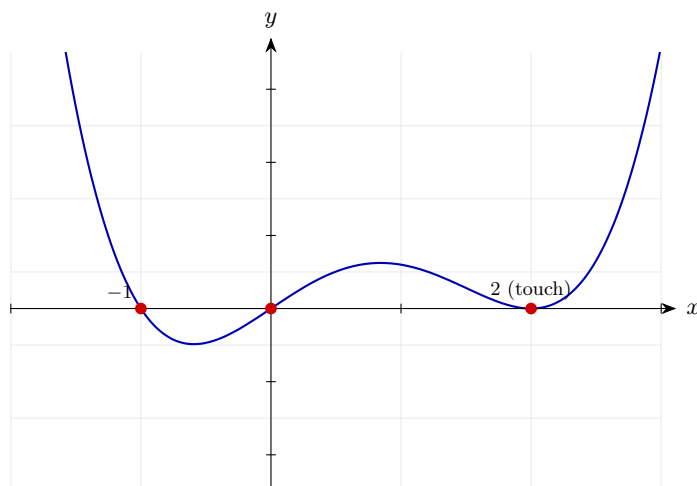
Step 4: Describe the sketch.

Crosses at $x = -1, 0$; touches at $x = 2$

The curve rises from bottom-left, crosses at -1 and 0 , bounces off $x = 2$, then rises to the right.

Final answer: Quartic crossing at $x = -1, 0$, touching at $x = 2$; y -intercept at origin; u-shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 21 (cg5-021)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = -x^2(x - 4)^2$, showing all intercepts and the nature of each root.

Worked solution.

Step 1: Identify the leading term.

$$-x^2 \cdot x^2 = -x^4 \quad (\text{negative})$$

Even degree, negative leading coefficient: n-shape (opens downward).

Step 2: Find the y-intercept.

$$y = -(0)^2(0 - 4)^2 = 0$$

Passes through the origin.

Step 3: Find the x-intercepts.

$$x = 0 \text{ (double, touches); } x = 4 \text{ (double, touches)}$$

Both roots are double: the curve touches the axis at both but does not cross.

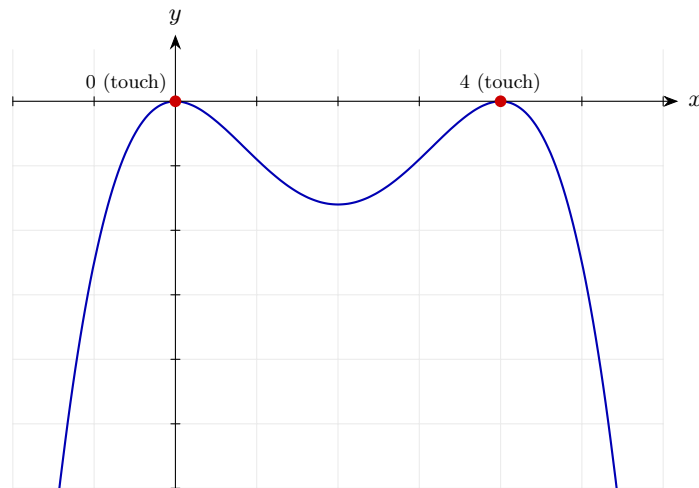
Step 4: Describe the sketch.

Touches at $x = 0$ and $x = 4$; lies below x -axis between them

The curve is entirely on or below the x -axis (since $y \leq 0$ for all x).

Final answer: Quartic touching x -axis at $x = 0$ and $x = 4$; curve lies on or below x -axis; n-shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 22 (cg5-022)

Difficulty: Foundation / Marks: 6

a) Factorise $x^4 - 5x^2 + 4$ completely.

b) Hence sketch the graph of $y = x^4 - 5x^2 + 4$, showing where it meets the axes.

Worked solution.

Step 1: Part a: let $u = x^2$.

$$u^2 - 5u + 4 = (u - 1)(u - 4)$$

Factors of 4 that add to -5 : -1 and -4 .

Step 2: Replace u with x^2 and factorise further.

$$(x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$$

Each quadratic factor is a difference of two squares.

Step 3: Part b: find the x -intercepts.

$$x = \pm 1, \pm 2$$

Four distinct roots, symmetric about the y -axis.

Step 4: Find the y -intercept.

$$y = 0 - 0 + 4 = 4$$

Substitute $x = 0$.

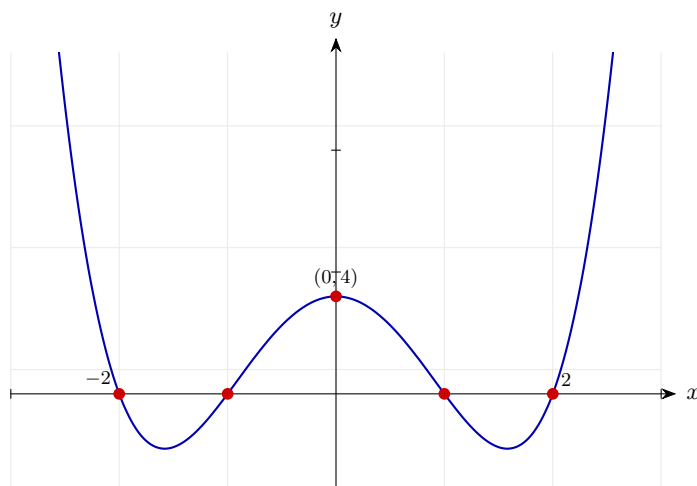
Step 5: Describe the shape.

x^4 has positive coefficient: u-shape overall

The curve rises from bottom-left and bottom-right, crossing at ± 1 and ± 2 .

Final answer: a) $(x - 1)(x + 1)(x - 2)(x + 2)$; b) Quartic crossing at $\pm 1, \pm 2$; y -intercept $(0, 4)$; u-shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 23 (cg5-023)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = \frac{4}{x}$, stating the equations of the asymptotes and the coordinates of the points where $x = 1$ and $x = -1$.

Worked solution.

Step 1: Identify the form: $y = kx^{-1}$ with $k = 4 > 0$ and odd power.

$$y = 4x^{-1}$$

Odd power, positive k : graph in top-right and bottom-left quadrants.

Step 2: State the asymptotes.

$$x = 0 \quad \text{and} \quad y = 0$$

Reciprocal functions never touch the axes — these are the asymptotes.

Step 3: Find the point where $x = 1$.

$$y = \frac{4}{1} = 4 \implies (1, 4)$$

Substitute $x = 1$.

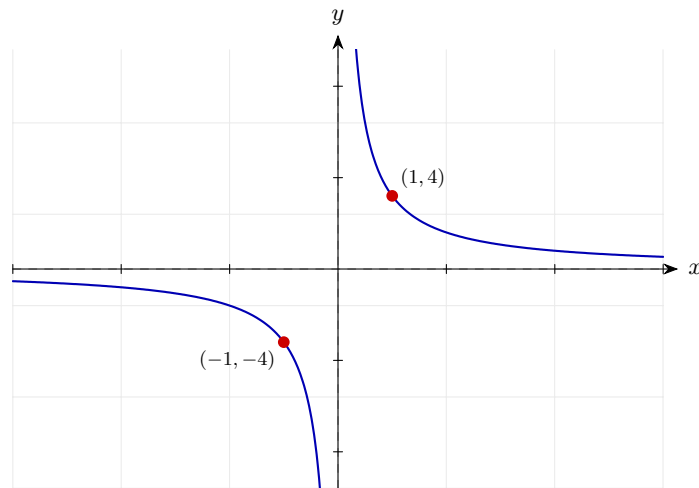
Step 4: Find the point where $x = -1$.

$$y = \frac{4}{-1} = -4 \implies (-1, -4)$$

By symmetry, $(-1, -4)$ lies on the lower-left branch.

Final answer: Asymptotes $x = 0, y = 0$; passes through $(1, 4)$ and $(-1, -4)$; two branches in top-right and bottom-left quadrants.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 24 (cg5-024)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = -\frac{5}{x}$, stating the asymptotes and the coordinates where $x = 1$ and $x = -1$.

Worked solution.

Step 1: Identify: $y = -5x^{-1}$, odd power, negative k .

$$k = -5 < 0, \quad n = 1 \text{ (odd)}$$

Odd power, negative k : graph in top-left and bottom-right quadrants.

Step 2: State the asymptotes.

$$x = 0 \quad \text{and} \quad y = 0$$

Same asymptotes as all basic reciprocal functions.

Step 3: Point at $x = 1$.

$$y = -\frac{5}{1} = -5 \implies (1, -5)$$

This lies in the bottom-right quadrant.

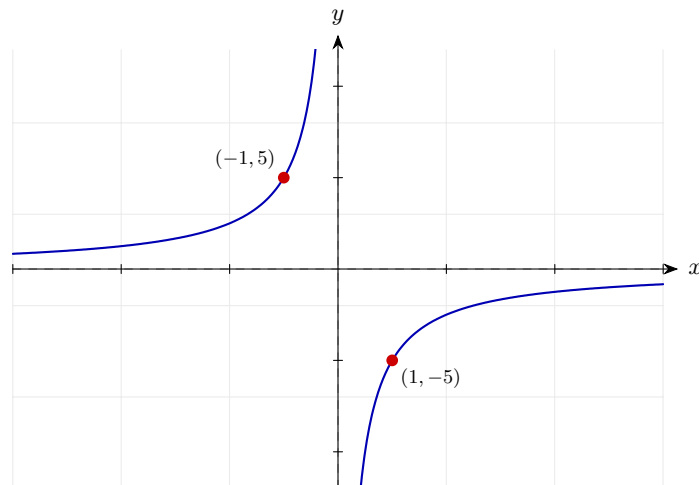
Step 4: Point at $x = -1$.

$$y = -\frac{5}{-1} = 5 \implies (-1, 5)$$

This lies in the top-left quadrant.

Final answer: Asymptotes $x = 0, y = 0$; passes through $(1, -5)$ and $(-1, 5)$; branches in top-left and bottom-right.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 25 (cg5-025)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = \frac{3}{x^2}$, stating the asymptotes and the points where $x = 1$ and $x = -1$.

Worked solution.

Step 1: Identify: $y = 3x^{-2}$, even power, positive k .

$$k = 3 > 0, \quad n = 2 \text{ (even)}$$

Even power, positive k : both branches above the x -axis.

Step 2: State the asymptotes.

$$x = 0 \quad \text{and} \quad y = 0$$

The curve approaches both axes but never touches them.

Step 3: Points at $x = \pm 1$.

$$y = \frac{3}{1} = 3 \implies (1, 3) \text{ and } (-1, 3)$$

Even power means x and $-x$ give the same y -value: the graph is symmetric about the y -axis.

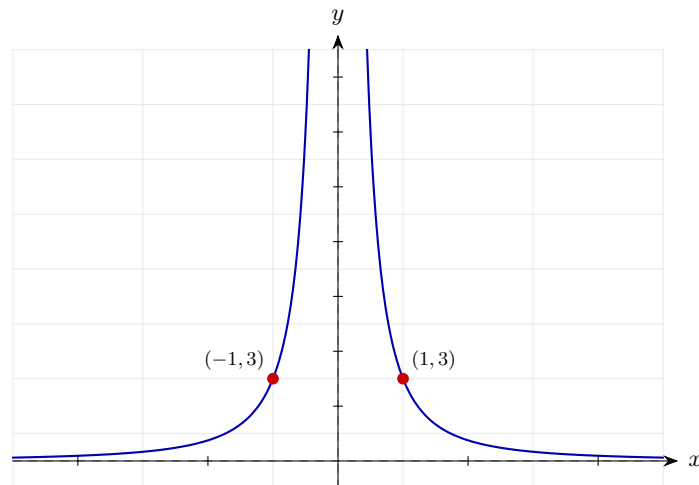
Step 4: Describe the sketch.

Two branches, both above x -axis, symmetric about y -axis

Both branches approach the asymptotes as $x \rightarrow 0$ and as $x \rightarrow \pm\infty$.

Final answer: Asymptotes $x = 0, y = 0$; passes through $(\pm 1, 3)$; both branches above x -axis; symmetric about y -axis.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 26 (cg5-026)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = -\frac{2}{x^4}$, stating the asymptotes and the points where $x = 1$ and $x = -1$.

Worked solution.

Step 1: Identify: $y = -2x^{-4}$, even power, negative k .

$$k = -2 < 0, \quad n = 4 \text{ (even)}$$

Even power, negative k : both branches below the x -axis.

Step 2: State the asymptotes.

$$x = 0 \quad \text{and} \quad y = 0$$

Standard asymptotes for all reciprocal functions of this type.

Step 3: Points at $x = \pm 1$.

$$y = -\frac{2}{1^4} = -2 \implies (1, -2) \text{ and } (-1, -2)$$

Both points at $y = -2$ by symmetry.

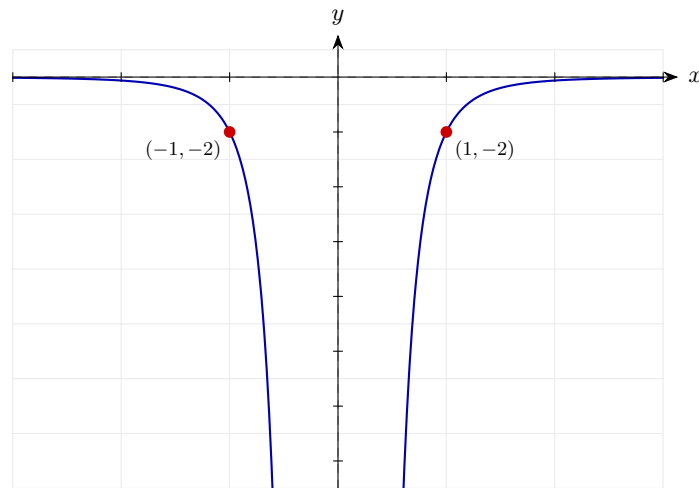
Step 4: Describe the sketch.

Two branches, both below x -axis, symmetric about y -axis

The entire graph lies below the x -axis.

Final answer: Asymptotes $x = 0, y = 0$; passes through $(\pm 1, -2)$; both branches below x -axis.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 27 (cg5-027)

Difficulty: Foundation / Marks: 4

Sketch the graphs of $y = \frac{6}{x}$ and $y = \frac{6}{x^2}$ on the same axes, for $x > 0$. State one key difference between the two graphs in the first quadrant.

Worked solution.

Step 1: Sketch $y = \frac{6}{x}$ for $x > 0$.

$$x = 1 \Rightarrow y = 6; \quad x = 2 \Rightarrow y = 3; \quad x = 6 \Rightarrow y = 1$$

This is a standard hyperbola branch in the first quadrant.

Step 2: Sketch $y = \frac{6}{x^2}$ for $x > 0$.

$$x = 1 \Rightarrow y = 6; \quad x = 2 \Rightarrow y = 1.5; \quad x = 3 \Rightarrow y = \frac{2}{3}$$

This curve falls more steeply because x^2 grows faster than x .

Step 3: Both pass through (1, 6); compare their rates of decrease.

$$\frac{6}{x^2} < \frac{6}{x} \text{ for } x > 1$$

The x^{-2} curve decreases more rapidly than the x^{-1} curve.

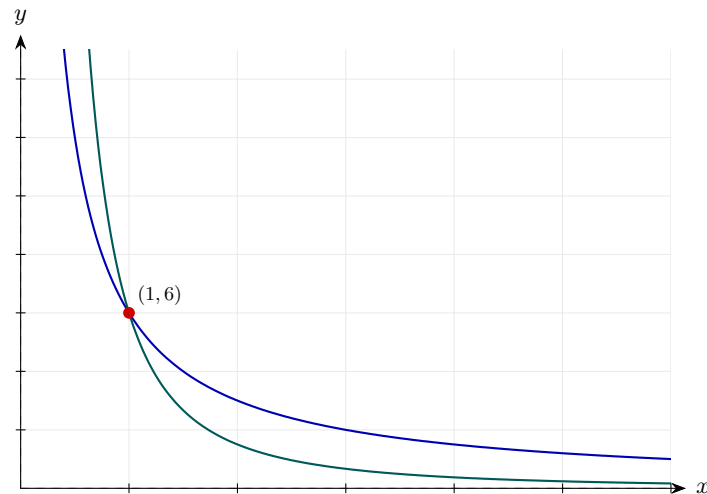
Step 4: State the key difference.

$$y = \frac{6}{x^2} \text{ falls more steeply than } y = \frac{6}{x} \text{ for } x > 1$$

Both have the same asymptotes, but the $1/x^2$ curve approaches zero faster.

Final answer: Both curves pass through (1, 6) and have asymptotes $x = 0, y = 0$. The graph of $y = \frac{6}{x^2}$ decreases more steeply for $x > 1$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 28 (cg5-028)

Difficulty: Foundation / Marks: 4

Match each function below to the correct description of its graph.

Functions: $y = 5x^{-3}$, $y = -\frac{1}{x^2}$, $y = \frac{7}{x^4}$, $y = -4x^{-5}$

Descriptions: - **P:** both branches above the x -axis, side by side - **Q:** top-left and bottom-right quadrant branches - **R:** top-right and bottom-left quadrant branches - **S:** both branches below the x -axis, side by side

Worked solution.

Step 1: $y = 5x^{-3}$: odd power (3), positive k .

Odd, positive $\Rightarrow$ top-right and bottom-left $\Rightarrow R$

Odd power with positive coefficient gives the standard hyperbola orientation.

Step 2: $y = -\frac{1}{x^2}$: even power (2), negative k .

Even, negative $\Rightarrow$ both branches below $\Rightarrow S$

Even power means side-by-side branches; negative means below the axis.

Step 3: $y = \frac{7}{x^4}$: even power (4), positive k .

Even, positive $\Rightarrow$ both branches above $\Rightarrow P$

Even power means side-by-side; positive means above the axis.

Step 4: $y = -4x^{-5}$: odd power (5), negative k .

Odd, negative $\Rightarrow$ top-left and bottom-right $\Rightarrow Q$

Odd power with negative coefficient flips the standard orientation.

Final answer: $5x^{-3} \rightarrow R$; $-1/x^2 \rightarrow S$; $7/x^4 \rightarrow P$; $-4x^{-5} \rightarrow Q$

Curve Sketching 29 (cg5-029)

Difficulty: Foundation / Marks: 5

a) Sketch the graphs of $y = \frac{2}{x^2}$ and $y = -x^2 + 3$ on the same axes.

b) Hence state the number of real solutions to $\frac{2}{x^2} = -x^2 + 3$.

Worked solution.

Step 1: Sketch $y = \frac{2}{x^2}$.

Even power, positive k : both branches above x -axis; asymptotes $x = 0, y = 0$

Passes through $(1, 2)$ and $(-1, 2)$.

Step 2: Sketch $y = -x^2 + 3$.

Downward parabola; vertex at $(0, 3)$; crosses x -axis at $x = \pm\sqrt{3}$

This is an inverted parabola with y -intercept 3.

Step 3: Solve the equation algebraically to confirm the count.

$$\frac{2}{x^2} = -x^2 + 3 \implies 2 = -x^4 + 3x^2 \implies x^4 - 3x^2 + 2 = 0$$

Multiply both sides by x^2 (valid since $x \neq 0$ — the reciprocal curve excludes this) and rearrange. This is a quadratic in x^2 .

Step 4: Factorise and solve.

$$(x^2 - 1)(x^2 - 2) = 0 \implies x = \pm 1, \pm\sqrt{2}$$

Each value of x^2 gives two real x -values, so there are four real solutions. Graphically, the parabola and the reciprocal curve cross twice on each side of the y -axis.

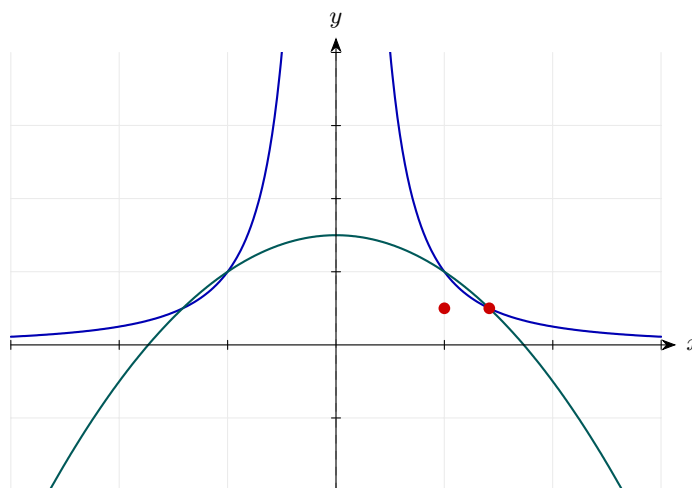
Step 5: State the conclusion.

$\Rightarrow$ 4 real solutions

Each intersection of the graphs corresponds to one real solution of the equation.

Final answer: The equation has 4 real solutions: $x = \pm 1, \pm\sqrt{2}$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 30 (cg5-030)

Difficulty: Foundation / Marks: 5

Sketch the graph of $y = (4-x)(x+1)^2$, showing all intercepts and describing the behaviour at each root.

Worked solution.

Step 1: Determine the leading term.

$$(4-x)(x+1)^2 \approx (-x)(x^2) = -x^3 \quad (\text{for large } |x|)$$

Negative leading coefficient: top-left to bottom-right shape.

Step 2: Find the y -intercept.

$$y = (4)(1)^2 = 4$$

Substitute $x = 0$.

Step 3: Find the x -intercepts and their nature.

$$4-x=0 \Rightarrow x=4 \text{ (crosses); } (x+1)^2=0 \Rightarrow x=-1 \text{ (double, touches)}$$

Single root at $x = 4$, double root at $x = -1$.

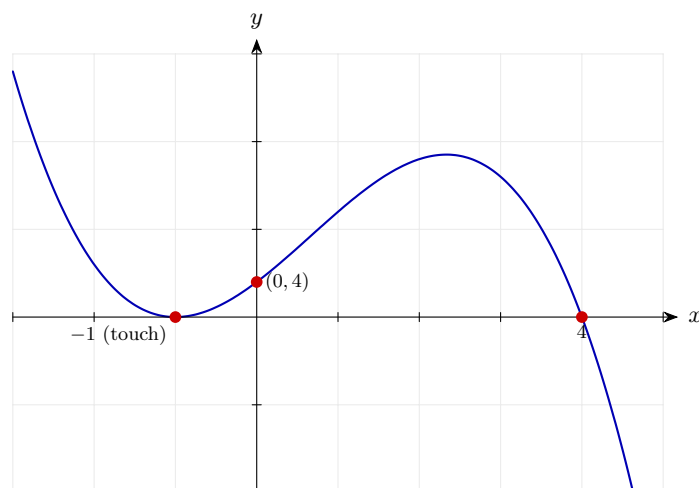
Step 4: Describe the sketch.

Enters top-left, touches $(-1, 0)$, passes through $(0, 4)$, crosses $(4, 0)$, falls bottom-right

The curve bounces off the axis at $x = -1$ and cuts through at $x = 4$.

Final answer: Cubic touching at $x = -1$, crossing at $x = 4$; y -intercept at $(0, 4)$; top-left to bottom-right shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 31 (cg5-031)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = x^4 - 16$, showing where it meets the coordinate axes.

Worked solution.

Step 1: Identify the shape.

x^4 with positive coefficient: u-shape

Even degree, positive leading coefficient.

Step 2: Find the y-intercept.

$$y = 0 - 16 = -16$$

Substitute $x = 0$.

Step 3: Find the x-intercepts by factorising.

$$x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

$x^2 + 4 = 0$ has no real solutions, so only real roots are $x = \pm 2$.

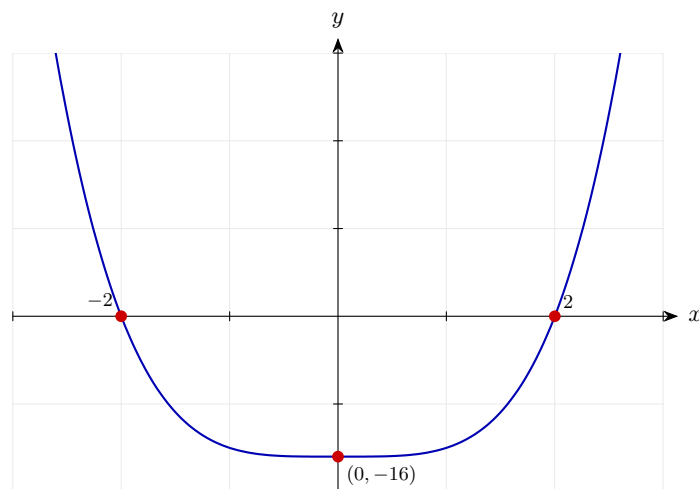
Step 4: Describe the sketch.

Crosses x -axis at ± 2 ; y -intercept at $(0, -16)$

The curve crosses the x -axis at two points, dips to -16 at the origin.

Final answer: Quartic crossing at $x = \pm 2$; y -intercept at $(0, -16)$; u-shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 32 (cg5-032)

Difficulty: Foundation / Marks: 7

- Factorise completely $y = x^3 - 4x^2 + 4x$.
- Sketch the graph, showing all intercepts and describing the nature of each root.
- Write down the number of distinct real roots of $x^3 - 4x^2 + 4x = 0$.

Worked solution.

Step 1: Part a: take out the common factor x .

$$x^3 - 4x^2 + 4x = x(x^2 - 4x + 4)$$

Every term contains x .

Step 2: Factorise the quadratic.

$$x^2 - 4x + 4 = (x - 2)^2$$

Perfect square trinomial: $(x - 2)^2 = x^2 - 4x + 4$.

Step 3: Write the full factorisation.

$$x(x - 2)^2$$

Single root at $x = 0$, double root at $x = 2$.

Step 4: Part b: find intercepts.

$$x = 0 \text{ (crosses); } x = 2 \text{ (double, touches); } y(0) = 0$$

Positive cubic shape (bottom-left to top-right).

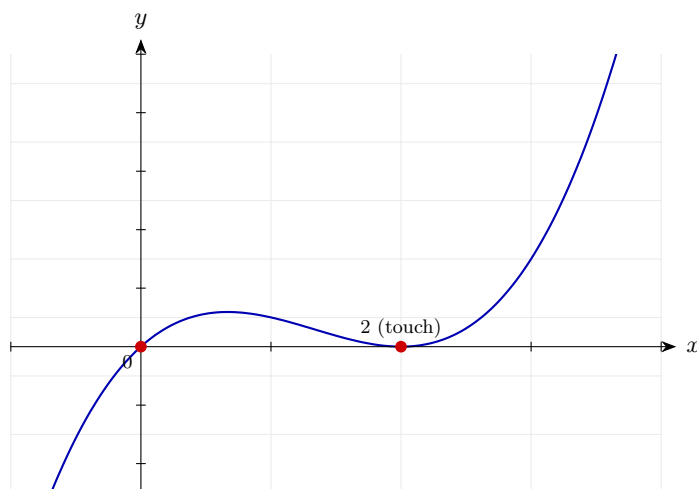
Step 5: Part c: count distinct real roots.

$$x = 0 \text{ (once), } x = 2 \text{ (once, even though doubled)}$$

There are 2 distinct real roots: $x = 0$ and $x = 2$.

Final answer: a) $x(x - 2)^2$; b) Crosses at $x = 0$, touches at $x = 2$; c) 2 distinct real roots.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 33 (cg5-033)

Difficulty: Foundation / Marks: 5

The graph of $y = f(x)$ is a quartic with positive leading coefficient. It crosses the x -axis at $x = -2$, touches the x -axis at $x = 1$, and crosses again at $x = 4$.

a) Write down a possible equation for $f(x)$ in factorised form.

b) Find the y -intercept of this graph.

Worked solution.

Step 1: Part a: use the root information to write the factors.

$$f(x) = (x + 2)(x - 1)^2(x - 4)$$

Cross at -2 and 4 gives single factors; touch at 1 gives a squared factor. The product of four linear factors gives degree 4 with positive leading coefficient.

Step 2: Check the leading coefficient.

$$x \cdot x^2 \cdot x = x^4 \quad (\text{positive})$$

Confirmed: positive leading coefficient.

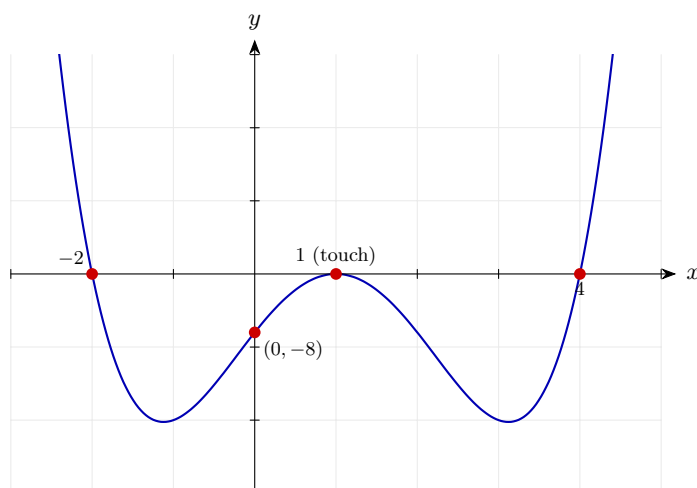
Step 3: Part b: find the y -intercept.

$$f(0) = (0 + 2)(0 - 1)^2(0 - 4) = 2 \times 1 \times (-4) = -8$$

Substitute $x = 0$ into the factorised equation.

Final answer: a) $f(x) = (x + 2)(x - 1)^2(x - 4)$; b) y -intercept at $(0, -8)$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 34 (cg5-034)

Difficulty: Foundation / Marks: 6

- a) On the same axes, sketch the graphs of $y = \frac{1}{x^3}$ and $y = x^3$.
- b) State the number of real solutions to $\frac{1}{x^3} = x^3$, and find them.

Worked solution.

Step 1: Part a: sketch $y = x^3$.

Odd power, positive: bottom-left to top-right; passes through origin

Standard cubic shape through $(-1, -1)$, $(0, 0)$, $(1, 1)$.

Step 2: Sketch $y = \frac{1}{x^3} = x^{-3}$.

Odd power, positive: top-right and bottom-left quadrants; asymptotes $x = 0, y = 0$

Passes through $(1, 1)$ and $(-1, -1)$.

Step 3: Part b: find intersections algebraically.

$$\frac{1}{x^3} = x^3 \implies x^6 = 1 \implies x = \pm 1$$

Multiply both sides by x^3 (note $x \neq 0$), then solve $x^6 = 1$.

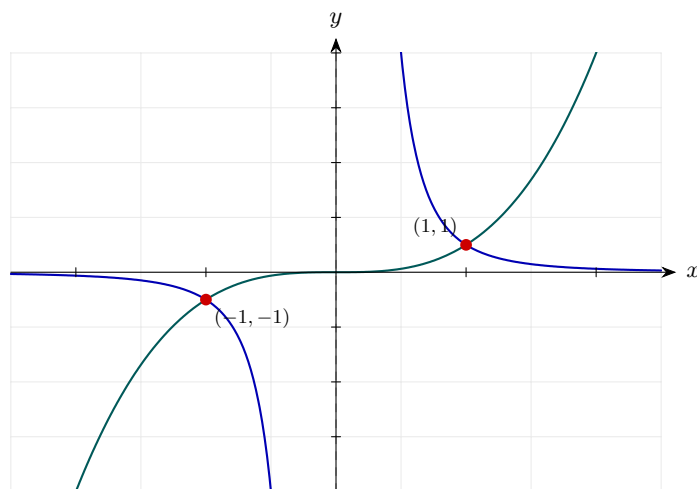
Step 4: Check: $x = 1$ gives $y = 1$; $x = -1$ gives $y = -1$.

Intersections at $(1, 1)$ and $(-1, -1)$

Both solutions confirmed.

Final answer: b) 2 real solutions: $x = 1$ and $x = -1$.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 35 (cg5-035)

Difficulty: Foundation / Marks: 8

The graph of $y = x^4 - 10x^3 + 35x^2 - 50x + 24$ has four positive integer roots.

a) By trying small positive integers, find all four roots.

b) Write the function in fully factorised form.

c) Sketch the graph, showing all intercepts.

Worked solution.

Step 1: Part a: try $x = 1$.

$$1 - 10 + 35 - 50 + 24 = 0 \checkmark$$

$x = 1$ is a root.

Step 2: Try $x = 2$.

$$16 - 80 + 140 - 100 + 24 = 0 \checkmark$$

$x = 2$ is a root.

Step 3: Try $x = 3$.

$$81 - 270 + 315 - 150 + 24 = 0 \checkmark$$

$x = 3$ is a root.

Step 4: By Vieta's formulas (sum of roots = 10), find the fourth root.

$$1 + 2 + 3 + r_4 = 10 \implies r_4 = 4$$

The sum of the roots of a monic quartic equals the negative of the x^3 coefficient: $-(-10) = 10$.

Step 5: Part b: write the factorised form.

$$y = (x - 1)(x - 2)(x - 3)(x - 4)$$

Verify by noting the product of roots $= 1 \times 2 \times 3 \times 4 = 24$ matches the constant term. $\checkmark$

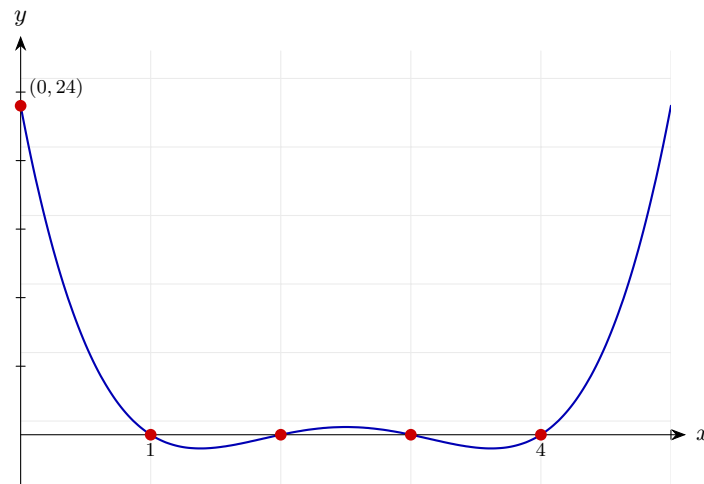
Step 6: Part c: find the y -intercept and describe the sketch.

$$y(0) = (-1)(-2)(-3)(-4) = 24$$

Positive leading coefficient (x^4), u-shape overall. Crosses at $x = 1, 2, 3, 4$; y -intercept at 24.

Final answer: a) Roots: 1, 2, 3, 4; b) $y = (x - 1)(x - 2)(x - 3)(x - 4)$; c) Quartic crossing at $x = 1, 2, 3, 4$; y -intercept at (0, 24); u-shape.

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 36 (cg5-036)

Difficulty: Foundation / Marks: 3

Sketch $y = x^2 - 4$, showing the intercepts with both axes.

Worked solution.

Step 1: y -intercept: set $x = 0$

$$y = -4$$

Substituting $x = 0$ into $y = x^2 - 4$ gives the y -axis crossing.

Step 2: x -intercepts: set $y = 0$

$$x^2 = 4 \implies x = \pm 2$$

Take square roots of both sides — remember the $\pm$ because both 2 and -2 square to 4.

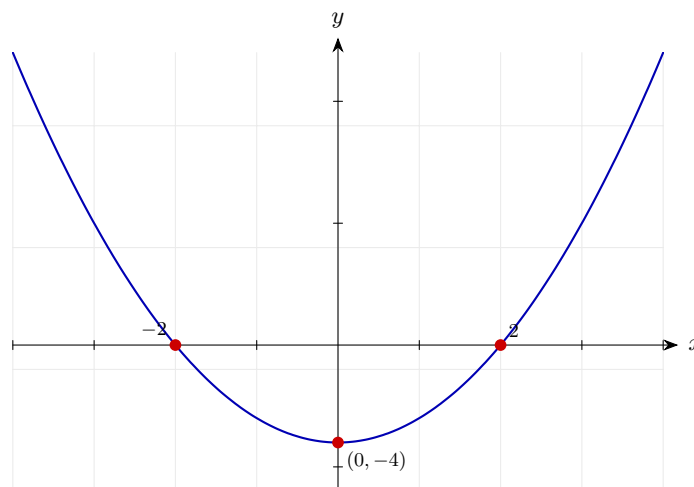
Step 3: Shape

U-shape, vertex at $(0, -4)$

Positive coefficient of x^2 gives a u-shape; the curve $y = x^2 - 4$ is $y = x^2$ shifted down by 4, so the vertex moves from $(0, 0)$ to $(0, -4)$.

Final answer: U-shape; roots at $(-2, 0)$ and $(2, 0)$; y -intercept $(0, -4)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 37 (cg5-037)

Difficulty: Foundation / Marks: 4

Sketch $y = (x - 1)(x + 3)$, showing all intercepts and the vertex.

Worked solution.

Step 1: x -intercepts

$$x = 1 \text{ and } x = -3$$

Setting $y = 0$, each factor must equal zero, giving the two roots directly from the factorised form.

Step 2: y -intercept

$$y = (-1)(3) = -3$$

Substitute $x = 0$ into the factorised form: $(0 - 1)(0 + 3) = (-1)(3)$.

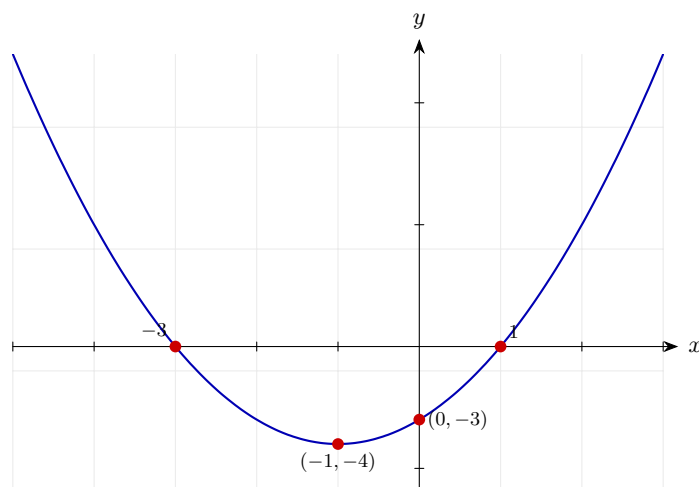
Step 3: Vertex (midpoint of roots)

$$x = \frac{1 + (-3)}{2} = -1; \quad y = (-2)(2) = -4$$

A parabola is symmetric, so its vertex sits at the midpoint of the two roots. Substituting $x = -1$ back into the factored form gives the y -coordinate.

Final answer: U-shape; roots $(1, 0)$, $(-3, 0)$; vertex $(-1, -4)$; y -int $(0, -3)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 38 (cg5-038)

Difficulty: Foundation / Marks: 4

Sketch $y = -x^2 + 6x - 5$, showing the vertex and all intercepts.

Worked solution.

Step 1: Factorise

$$y = -(x - 1)(x - 5)$$

Factor out -1 to get $-(x^2 - 6x + 5)$, then factorise the bracket: numbers multiplying to 5 and adding to -6 are -1 and -5 .

Step 2: Vertex

$$x = 3; \quad y = -9 + 18 - 5 = 4$$

Negative coefficient of x^2 means the parabola opens downwards, so the vertex is the maximum. It sits on the axis of symmetry midway between the roots: $x = \frac{1+5}{2} = 3$.

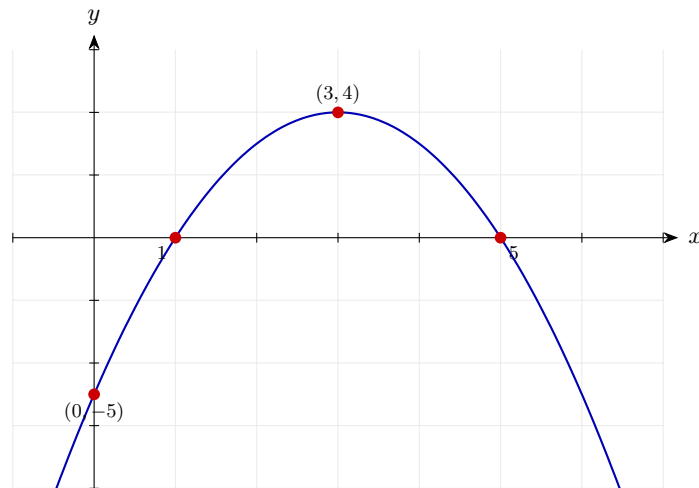
Step 3: y -intercept

$$y = -5$$

Substitute $x = 0$ into $y = -x^2 + 6x - 5$ to get the constant term -5 .

Final answer: n-shape; roots $(1, 0)$, $(5, 0)$; vertex $(3, 4)$; y -int $(0, -5)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 39 (cg5-039)

Difficulty: Foundation / Marks: 3

Sketch $y = \frac{1}{x}$ for $x > 0$ and $x < 0$, showing the asymptotes.

Worked solution.

Step 1: Asymptotes

$$x = 0 \text{ (vertical), } y = 0 \text{ (horizontal)}$$

The function is undefined at $x = 0$ (vertical asymptote), and as $|x| \rightarrow \infty$ the value $\frac{1}{x} \rightarrow 0$, giving the horizontal asymptote.

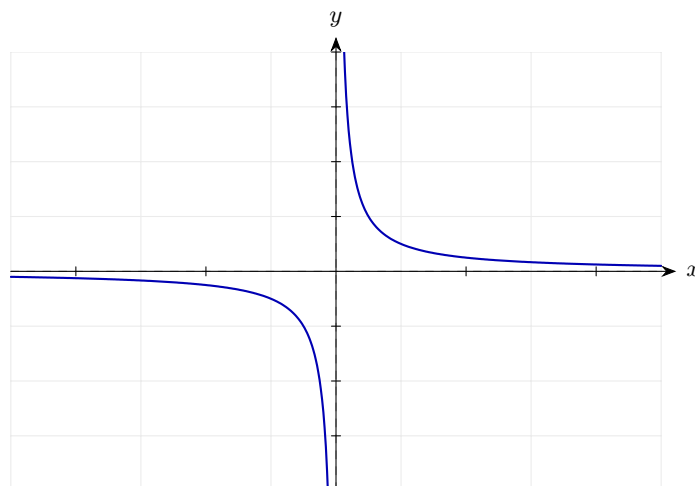
Step 2: Behaviour

$$x > 0 : \text{curve in 1st quadrant; } x < 0 : \text{curve in 3rd quadrant}$$

Positive x gives positive y and negative x gives negative y , so the two branches lie in opposite (diagonal) quadrants.

Final answer: Rectangular hyperbola with asymptotes $x = 0$ and $y = 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 40 (cg5-040)

Difficulty: Foundation / Marks: 2

Sketch $y = x^3$, showing the shape and any key points.

Worked solution.

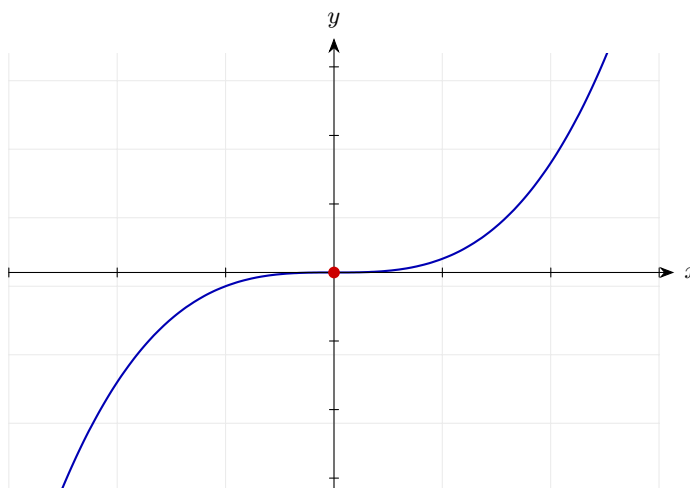
Key features

Through origin; point of inflection at $(0,0)$

S-shaped curve, negative for $x < 0$, positive for $x > 0$.

Final answer: S-shaped cubic through the origin

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 41 (cg5-041)

Difficulty: Foundation / Marks: 5

Sketch $y = x^3 - 3x$, showing all stationary points and intercepts.

Worked solution.

Step 1: Intercepts: $y = x(x^2 - 3)$

$$x = 0, \pm\sqrt{3}$$

Factor out x then set each factor to zero. The quadratic $x^2 - 3 = 0$ gives $x = \pm\sqrt{3}$.

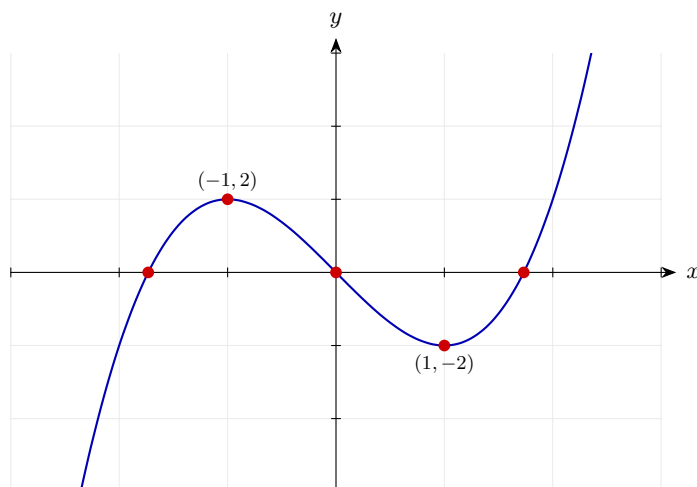
Step 2: Stationary points: $dy/dx = 3x^2 - 3 = 0$

$$x = \pm 1; \quad (1, -2) \text{ and } (-1, 2)$$

Differentiate and set $\frac{dy}{dx} = 0$. Substitute $x = 1$ into $y = x^3 - 3x$ to get $y = -2$ (minimum), and $x = -1$ to get $y = 2$ (maximum). Positive cubic, so the left turning point is the maximum.

Final answer: Cubic; roots at $0, \pm\sqrt{3}$; max $(-1, 2)$; min $(1, -2)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 42 (cg5-042)

Difficulty: Foundation / Marks: 3

Sketch $y = \frac{1}{x-2}$, clearly showing the asymptotes.

Worked solution.

Step 1: Asymptotes

$$x = 2 \text{ (vertical), } y = 0 \text{ (horizontal)}$$

This is $y = \frac{1}{x}$ shifted 2 units right (replacing x with $x - 2$), so the vertical asymptote moves from $x = 0$ to $x = 2$. The horizontal asymptote $y = 0$ is unchanged.

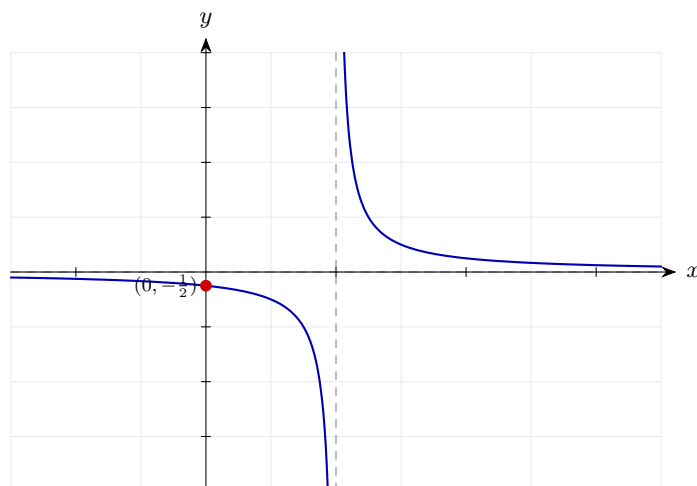
Step 2: y-intercept

$$y = \frac{1}{-2} = -\frac{1}{2}$$

Substitute $x = 0$ into $\frac{1}{x-2}$ to get the crossing on the y -axis.

Final answer: Translated reciprocal; asymptotes $x = 2, y = 0$; y -int $(0, -1/2)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 43 (cg5-043)

Difficulty: Foundation / Marks: 4

Sketch $y = (x - 2)^2(x + 1)$, showing all intercepts.

Worked solution.

Step 1: x -intercepts

$$x = 2 \text{ (repeated), } x = -1$$

Setting each factor to zero: $(x - 2)^2 = 0$ gives a double root $x = 2$ (so the curve touches the axis there), and $x + 1 = 0$ gives a single root $x = -1$ (the curve crosses).

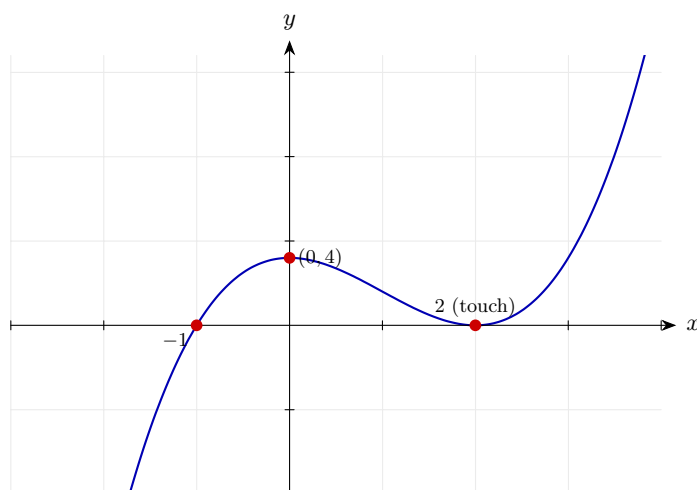
Step 2: y -intercept

$$y = (0 - 2)^2(0 + 1) = 4 \times 1 = 4$$

Substitute $x = 0$ into each factor: $(0 - 2)^2 = 4$ and $(0 + 1) = 1$, giving $y = 4$.

Final answer: Cubic; touches x -axis at $(2, 0)$; crosses at $(-1, 0)$; y -int $(0, 4)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 44 (cg5-044)

Difficulty: Foundation / Marks: 2

Sketch $y = \sqrt{x}$ for $x \geq 0$.

Worked solution.

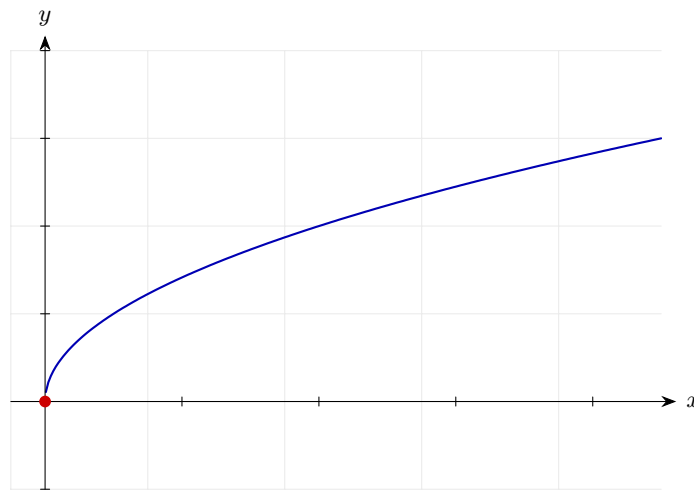
Key features

Starts at origin, increases, domain $x \geq 0$

Square roots are only defined for non-negative inputs, so the domain is $x \geq 0$. The curve starts at $(0, 0)$, passes through $(1, 1)$ and $(4, 2)$, and gets less steep as x grows (concave, gradient $\frac{1}{2\sqrt{x}}$ decreases).

Final answer: Starts at $(0, 0)$, increases and flattens; defined for $x \geq 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 45 (cg5-045)

Difficulty: Foundation / Marks: 3

Sketch $y = 2^x$, showing the asymptote and y-intercept.

Worked solution.

Step 1: y-intercept

$$y = 2^0 = 1$$

Any non-zero base raised to the power 0 is 1, so every exponential curve $y = a^x$ passes through $(0, 1)$.

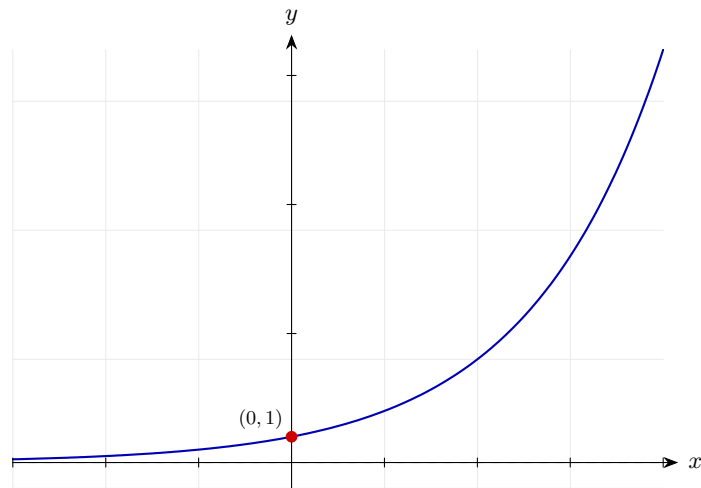
Step 2: Asymptote

$$y = 0 \text{ (horizontal, as } x \rightarrow -\infty \text{)}$$

As $x \rightarrow -\infty$, $2^x \rightarrow 0$ but never reaches it, so $y = 0$ is a horizontal asymptote. As $x \rightarrow +\infty$, 2^x grows without bound.

Final answer: Exponential curve; y-int $(0, 1)$; asymptote $y = 0$; increases rapidly

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 46 (cg5-046)

Difficulty: Foundation / Marks: 4

Sketch $y = x^2 + 2x + 5$, explaining why it does not cross the x -axis.

Worked solution.

Step 1: Complete the square

$$y = (x + 1)^2 + 4$$

Half the coefficient of x is 1, so $x^2 + 2x = (x + 1)^2 - 1$; adding 5 gives $(x + 1)^2 + 4$. Since $(x + 1)^2 \geq 0$ always, the minimum value of y is 4, reached at $x = -1$.

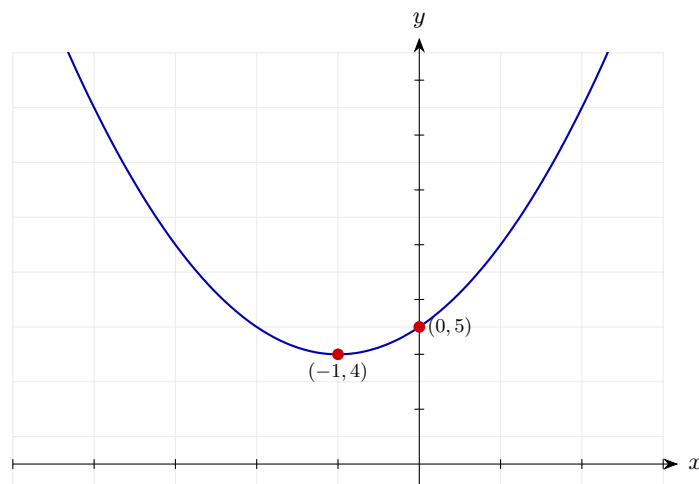
Step 2: Discriminant

$$\Delta = 4 - 20 = -16 < 0$$

Using $b^2 - 4ac$ with $a = 1, b = 2, c = 5$. A negative discriminant means no real roots, confirming the curve sits entirely above the x -axis.

Final answer: U-shape; vertex $(-1, 4)$; y -int $(0, 5)$; no x -intercepts

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 47 (cg5-047)

Difficulty: Foundation / Marks: 3

Sketch $y = |x - 3|$.

Worked solution.

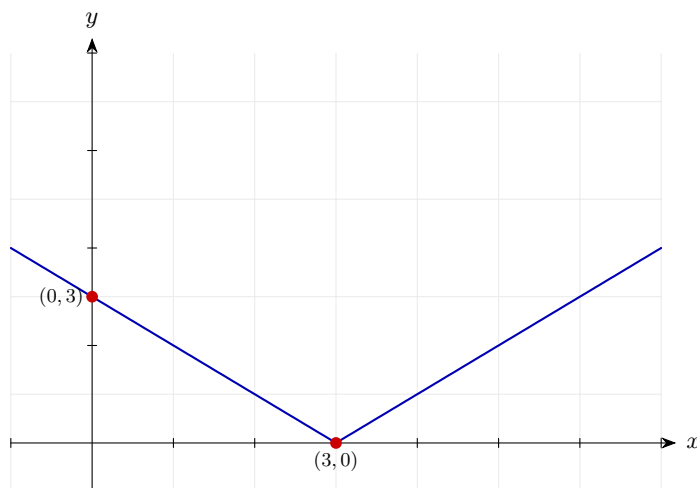
V-shape with vertex at $x = 3$

$$\text{Vertex at } (3, 0); \quad y\text{-int: } |0 - 3| = 3$$

The modulus reflects any negative output to positive, so $y = |x - 3|$ is $y = x - 3$ for $x \geq 3$ and $y = -(x - 3)$ for $x < 3$. The vertex sits where the inside equals zero ($x = 3$), giving a V-shape with gradient ± 1 .

Final answer: V-shape; vertex $(3, 0)$; y -int $(0, 3)$; symmetric about $x = 3$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 48 (cg5-048)

Difficulty: Foundation / Marks: 5

Sketch $y = x^4 - 4x^2$, showing all intercepts and stationary points.

Worked solution.

Step 1: Factorise

$$y = x^2(x^2 - 4) = x^2(x - 2)(x + 2)$$

Take out the common factor x^2 , then apply the difference of two squares to $x^2 - 4$. The double root at $x = 0$ means the curve touches the axis there, while $x = \pm 2$ are single crossings.

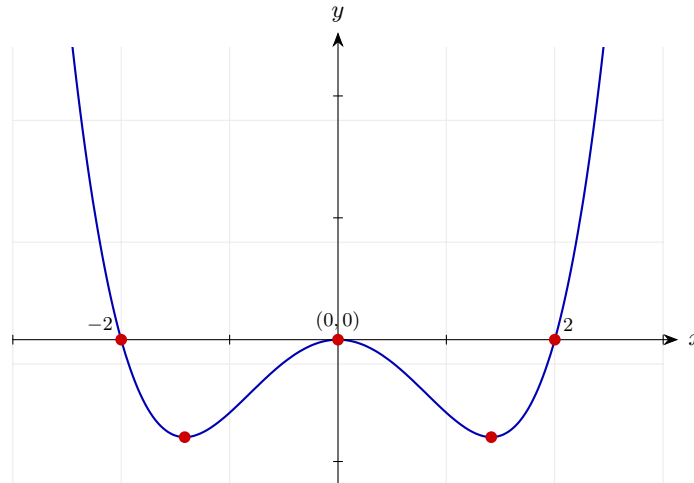
Step 2: Stationary points: $dy/dx = 4x^3 - 8x = 4x(x^2 - 2) = 0$

$$x = 0, \pm\sqrt{2}; \quad y(0) = 0, \quad y(\pm\sqrt{2}) = 4 - 8 = -4$$

Differentiating and factoring gives three stationary points. At $x = \pm\sqrt{2}$: $y = (\sqrt{2})^4 - 4(\sqrt{2})^2 = 4 - 8 = -4$. Since the leading coefficient is positive (u-shape overall), the outer turning points are minima and the central one at the origin is a local maximum, giving a W-shape.

Final answer: W-shape; roots at $0, \pm 2$; local max $(0, 0)$; local min $(\pm\sqrt{2}, -4)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 49 (cg5-049)

Difficulty: Foundation / Marks: 4

Sketch $y = \frac{2x}{x+1}$, showing the asymptotes and intercepts.

Worked solution.

Step 1: Asymptotes

$$x = -1 \text{ (vertical); } y = 2 \text{ (horizontal, since } \frac{2x}{x} \rightarrow 2)$$

The denominator is zero when $x = -1$, giving the vertical asymptote. As $|x| \rightarrow \infty$, the lower-order terms become negligible and the ratio tends to $\frac{2x}{x} = 2$.

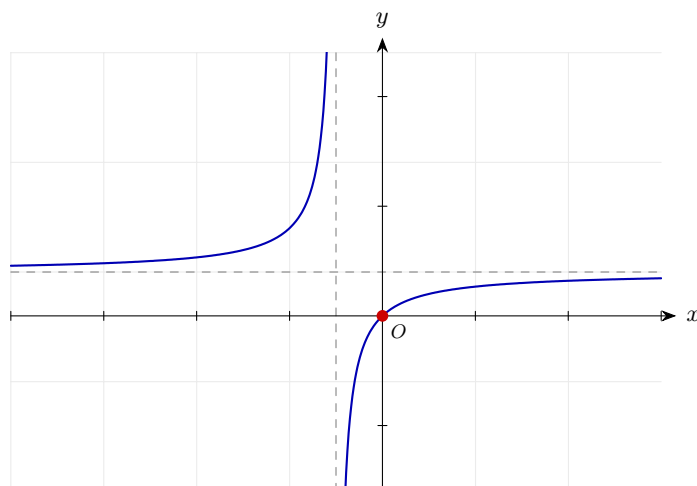
Step 2: Intercepts

$$x = 0 : y = 0; \quad y = 0 : x = 0$$

Substituting $x = 0$ gives $y = 0$; setting $y = 0$ requires $2x = 0$, so $x = 0$. Both intercepts coincide at the origin.

Final answer: Through origin; asymptotes $x = -1$ and $y = 2$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 50 (cg5-050)

Difficulty: Foundation / *Marks: 3*

The graph of $y = f(x)$ passes through $(0, 3)$ and has a maximum at $(2, 7)$. Sketch $y = f(x) + 2$.

Worked solution.

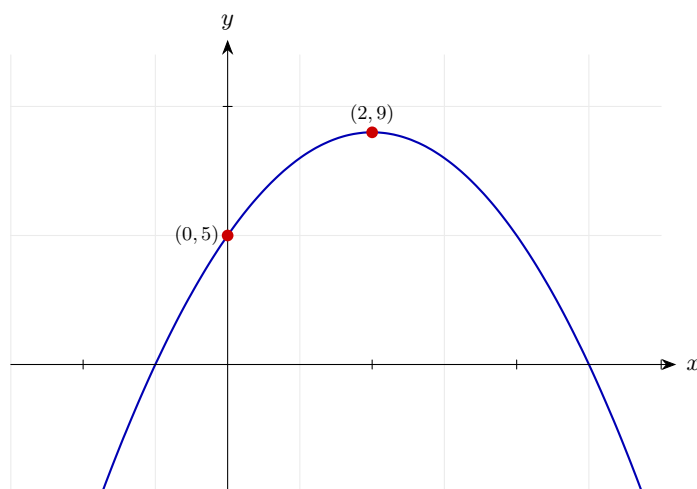
Vertical translation up 2

$$(0, 3) \rightarrow (0, 5); \quad (2, 7) \rightarrow (2, 9)$$

Replacing $f(x)$ with $f(x) + 2$ shifts the entire graph upwards by 2 units, so every point keeps its x -coordinate and has 2 added to its y -coordinate. The shape and the location of the maximum (relative to the curve) are unchanged.

Final answer: Same shape, shifted up 2; passes through $(0, 5)$; max at $(2, 9)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 51 (cg5-051)

Difficulty: Foundation / Marks: 3

Sketch $y = (x + 2)^3$, stating the point of inflection.

Worked solution.

Step 1: Translation of $y = x^3$

Point of inflection at $(-2, 0)$

Replacing x with $x + 2$ shifts the graph 2 units to the left. The inflection point of $y = x^3$ at the origin therefore moves to $(-2, 0)$, and this is also a triple root (the curve flattens then crosses the x -axis).

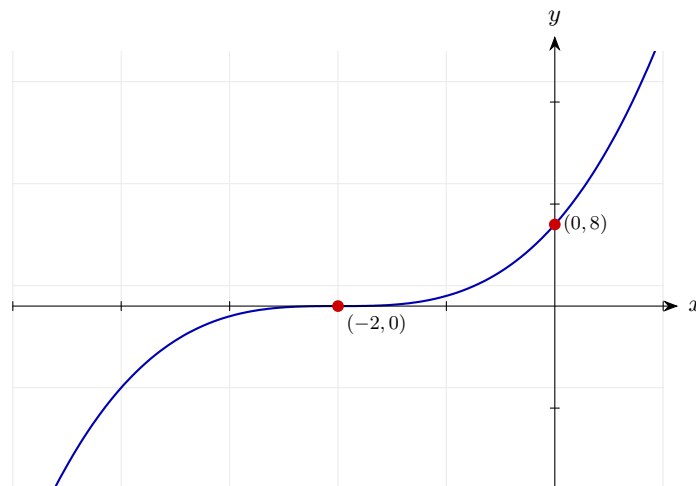
Step 2: y -intercept

$$y = 8$$

Substitute $x = 0$: $(0 + 2)^3 = 8$.

Final answer: Cubic S-shape; inflection $(-2, 0)$; y -int $(0, 8)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 52 (cg5-052)

Difficulty: Foundation / Marks: 2

Sketch $y = -\frac{1}{x}$, showing the asymptotes.

Worked solution.

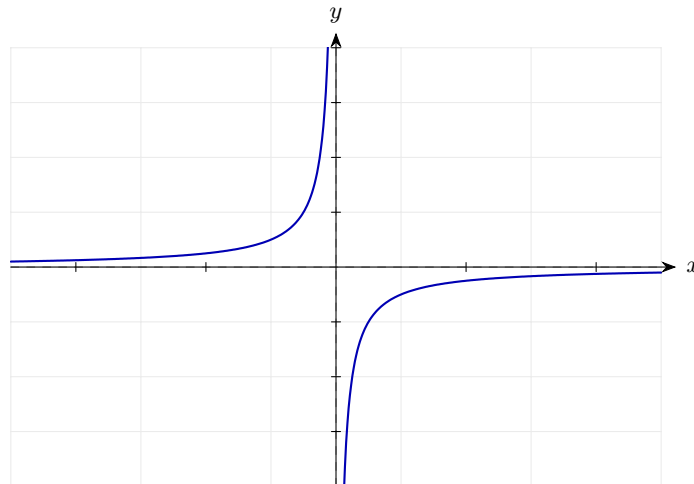
Reflection of $y = 1/x$ in x -axis

Asymptotes: $x = 0, y = 0$; curves in 2nd and 4th quadrants

Multiplying by -1 reflects $y = \frac{1}{x}$ in the x -axis, flipping each branch. Positive x now gives negative y (4th quadrant) and negative x gives positive y (2nd quadrant). The asymptotes $x = 0$ and $y = 0$ are unchanged.

Final answer: Hyperbola in 2nd and 4th quadrants; asymptotes $x = 0, y = 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 53 (cg5-053)

Difficulty: Foundation / Marks: 3

Sketch $y = x^2 - 6x + 9$, identifying the nature of the root.

Worked solution.

Step 1: Factorise

$$y = (x - 3)^2$$

This is a perfect square trinomial: $(x - 3)^2 = x^2 - 6x + 9$. The repeated factor means $x = 3$ is a double root.

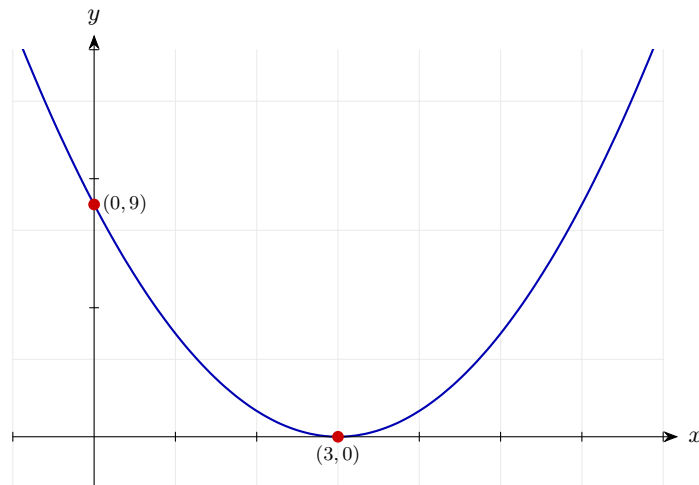
Step 2: Key features

Touches x-axis at $(3, 0)$; y-int $(0, 9)$

A double root means the curve touches but does not cross the x -axis at $x = 3$. The vertex coincides with this touch point, and substituting $x = 0$ gives $y = 9$.

Final answer: U-shape; touches x -axis at $(3, 0)$; vertex $(3, 0)$; y-int $(0, 9)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 54 (cg5-054)

Difficulty: Foundation / Marks: 3

Sketch $y = \frac{1}{x^2}$, showing the asymptotes.

Worked solution.

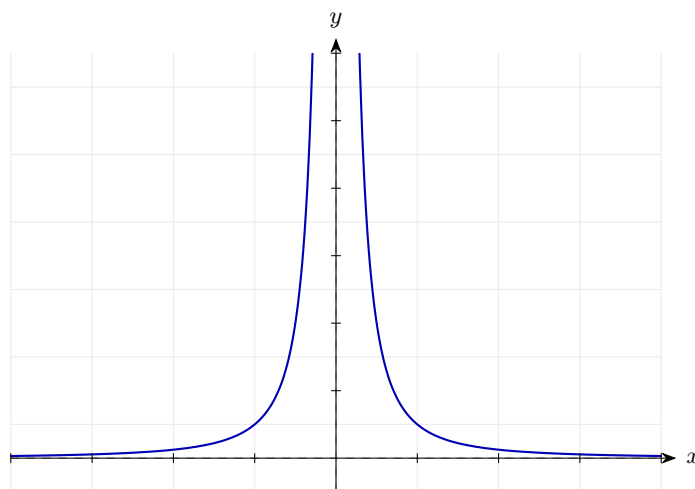
Key features

Always positive; asymptotes $x = 0$ and $y = 0$

Since $x^2 > 0$ for all $x \neq 0$, the output $\frac{1}{x^2}$ is always positive — both branches sit above the x -axis. Because $(-x)^2 = x^2$, the graph is symmetric about the y -axis, and the asymptotes are the two axes.

Final answer: Both branches in 1st and 2nd quadrants; asymptotes $x = 0, y = 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 55 (cg5-055)

Difficulty: Foundation / Marks: 3

Given $y = f(x)$ with roots at -1, 2, 5 and y-intercept 10, sketch $y = f(x + 1)$.

Worked solution.

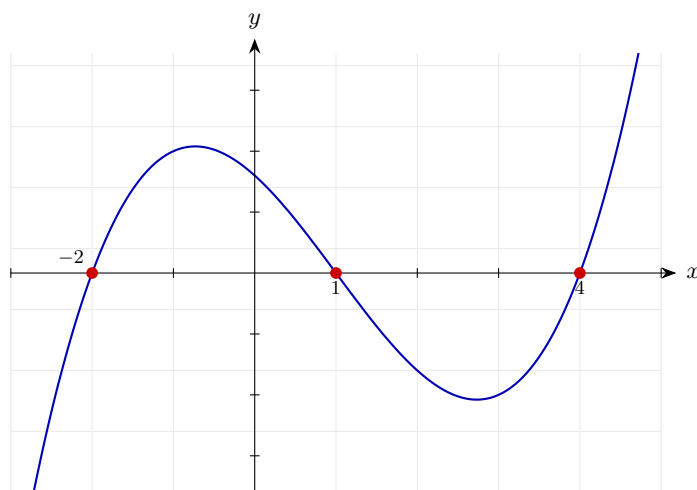
Shift left by 1

New roots: -2, 1, 4; new y-intercept = $f(1)$

Replacing x with $x + 1$ shifts the graph 1 unit to the left, so each root drops by 1: $-1 \rightarrow -2$, $2 \rightarrow 1$, $5 \rightarrow 4$. The new y-intercept is the value of the original function at $x = 1$ (unknown from the data given — only that $f(0) = 10$).

Final answer: Same shape shifted 1 left; new roots at -2, 1, 4; new y-intercept = $f(1)$ (not determined by data given)

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 56 (cg5-056)

Difficulty: Foundation / Marks: 3

Sketch $y = 3 - x^2$, showing the vertex and intercepts.

Worked solution.

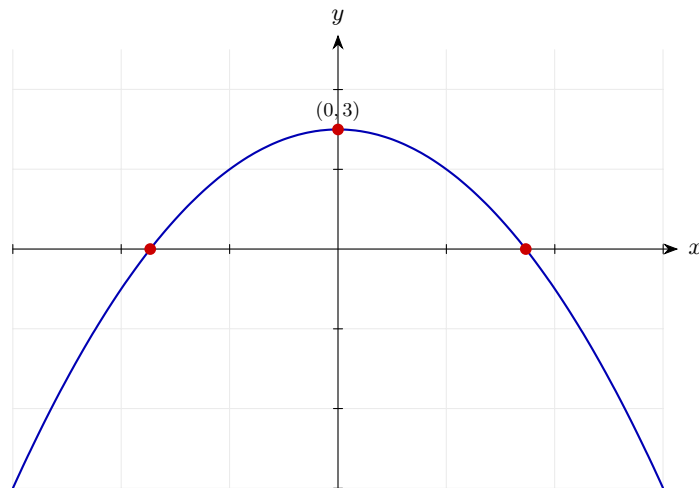
Key features

n-shape; vertex $(0, 3)$; $x = \pm\sqrt{3}$

Rewriting as $y = -x^2 + 3$: the negative leading coefficient gives an n-shape (opens downward), and the constant $+3$ shifts $y = -x^2$ up by 3 so the maximum sits at $(0, 3)$. Setting $y = 0$ gives $x^2 = 3 \Rightarrow x = \pm\sqrt{3}$.

Final answer: n-shape; vertex $(0, 3)$; roots $(\pm\sqrt{3}, 0)$; y-int $(0, 3)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 57 (cg5-057)

Difficulty: Foundation / Marks: 4

Sketch $y = x(x - 2)(x + 3)$, **showing all intercepts.**

Worked solution.

Step 1: x-intercepts

$$x = 0, 2, -3$$

Set each factor of the factorised cubic to zero. Three distinct roots mean the curve crosses the x -axis three times.

Step 2: y-intercept

$$y = 0$$

Substituting $x = 0$ into $x(x - 2)(x + 3)$ gives 0, so the curve passes through the origin.

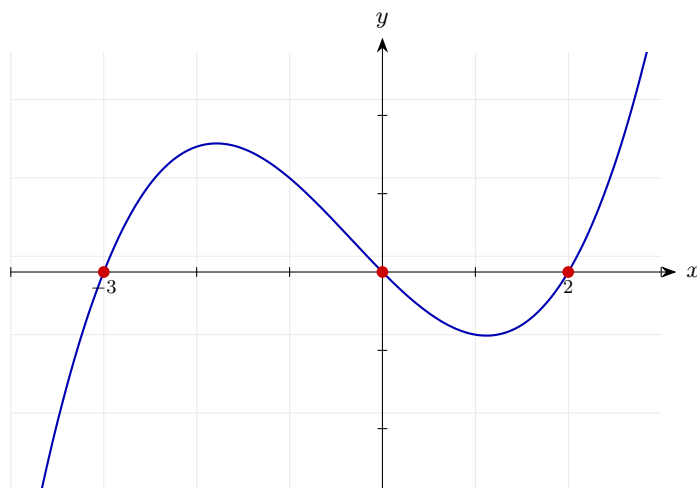
Step 3: End behaviour

$$x \rightarrow +\infty : y \rightarrow +\infty; \quad x \rightarrow -\infty : y \rightarrow -\infty$$

Expanding gives leading term $+x^3$. Positive coefficient of an odd power means bottom-left to top-right overall shape.

Final answer: Cubic through $(-3, 0)$, $(0, 0)$, $(2, 0)$; positive leading coefficient

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 58 (cg5-058)

Difficulty: Foundation / *Marks: 4*

Sketch $y = \frac{x+1}{x-1}$, showing the asymptotes and intercepts.

Worked solution.

Step 1: Asymptotes

$$x = 1 \text{ (vertical); } y = 1 \text{ (horizontal)}$$

The denominator is zero at $x = 1$, giving a vertical asymptote. For large $|x|$ the ratio behaves like $\frac{x}{x} = 1$, giving the horizontal asymptote $y = 1$.

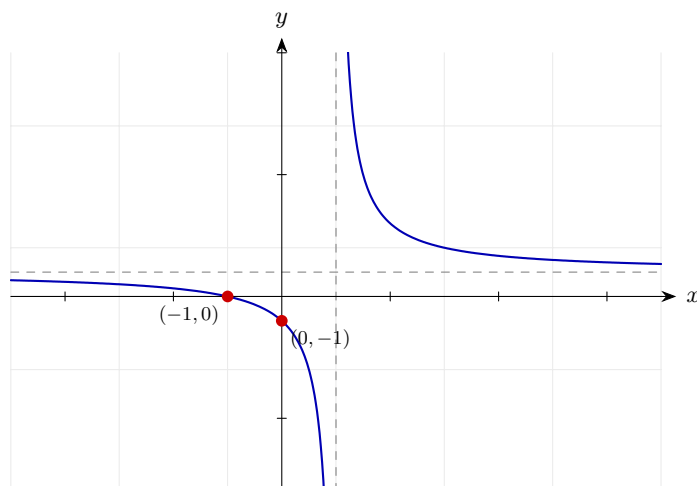
Step 2: Intercepts

$$x = 0 : y = -1; \quad y = 0 : x = -1$$

At $x = 0$: $y = \frac{0+1}{0-1} = -1$. Setting the numerator equal to zero (the only way the fraction is zero) gives $x + 1 = 0 \Rightarrow x = -1$.

Final answer: Asymptotes $x = 1, y = 1$; intercepts $(-1, 0), (0, -1)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 59 (cg5-059)

Difficulty: Foundation / Marks: 3

The curve $y = f(x)$ has a minimum at $(1, -3)$ and passes through $(0, 0)$. Sketch $y = -f(x)$.

Worked solution.

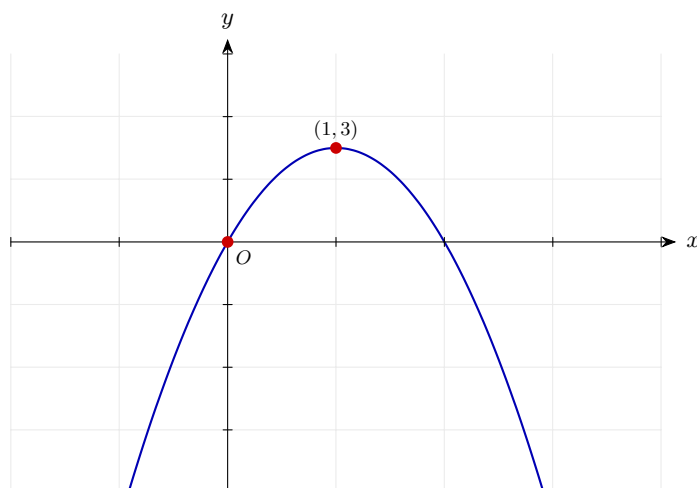
Reflect in x -axis

$$(1, -3) \rightarrow (1, 3); \quad (0, 0) \rightarrow (0, 0)$$

The transformation $y = -f(x)$ multiplies every y -value by -1 , which reflects the curve in the x -axis. x -coordinates stay the same, so the minimum at $(1, -3)$ becomes a maximum at $(1, 3)$, and the origin (where $y = 0$) maps to itself.

Final answer: Reflected curve; max at $(1, 3)$; through origin

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 60 (cg5-060)

Difficulty: Foundation / Marks: 2

Sketch $y = e^{-x}$, showing the asymptote and intercept.

Worked solution.

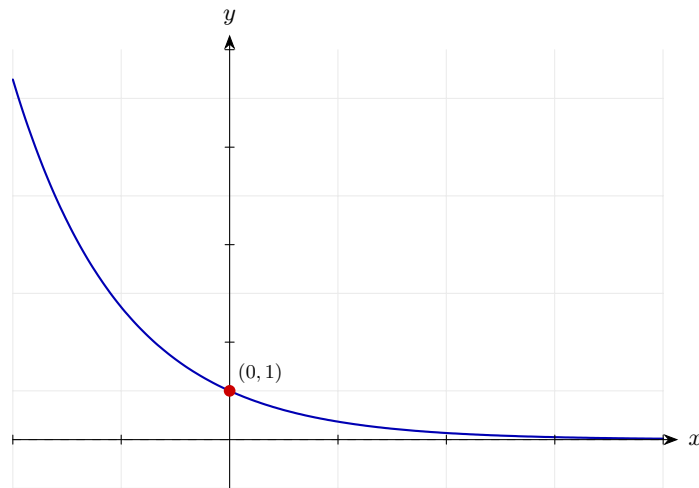
Key features

$$y\text{-int: } (0, 1); \quad \text{asymptote } y = 0; \quad \text{decreasing}$$

$y = e^{-x}$ is $y = e^x$ reflected in the y -axis (because x is replaced by $-x$), so it decreases instead of increases. At $x = 0$, $y = 1$; as $x \rightarrow +\infty$, $e^{-x} \rightarrow 0$ but never reaches it, giving the asymptote $y = 0$.

Final answer: Decreasing exponential; y -int $(0, 1)$; asymptote $y = 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 61 (cg5-061)

Difficulty: Foundation / Marks: 4

Sketch $y = x^2(x - 3)$, showing where the curve touches or crosses the x-axis.

Worked solution.

Step 1: Roots

$$x = 0 \text{ (repeated - touches), } x = 3 \text{ (crosses)}$$

Setting $y = 0$: the factor x^2 gives a double root at $x = 0$ (curve touches but does not cross), and $x - 3 = 0$ gives a single root $x = 3$ (curve cuts through).

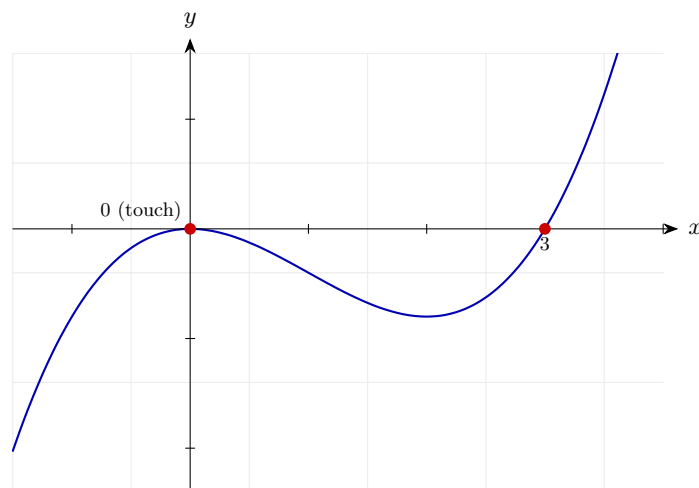
Step 2: y-intercept

$$y = 0$$

Substituting $x = 0$ gives $0^2 \cdot (-3) = 0$, so the curve passes through the origin.

Final answer: Cubic; touches x -axis at origin, crosses at $(3, 0)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 62 (cg5-062)

Difficulty: Foundation / Marks: 4

Sketch $y = \frac{4}{x+2} + 1$, showing the asymptotes and intercepts.

Worked solution.

Step 1: Asymptotes

$$x = -2 \text{ (vertical); } y = 1 \text{ (horizontal)}$$

This is $y = \frac{4}{x}$ shifted 2 left (replace x with $x + 2$) and 1 up. Each shift carries its asymptote: vertical $x = 0$ moves to $x = -2$, horizontal $y = 0$ moves to $y = 1$.

Step 2: y-intercept

$$y = \frac{4}{2} + 1 = 3$$

Substitute $x = 0$ into $\frac{4}{x+2} + 1$.

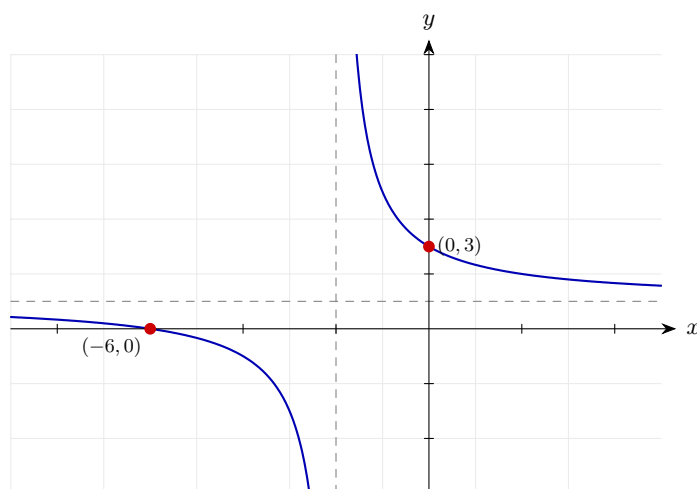
Step 3: x-intercept

$$0 = \frac{4}{x+2} + 1 \implies x = -6$$

Rearrange: $\frac{4}{x+2} = -1$, then multiply both sides by $x + 2$ to get $4 = -(x + 2)$, so $x = -6$.

Final answer: Translated reciprocal; asymptotes $x = -2, y = 1$; intercepts $(-6, 0), (0, 3)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 63 (cg5-063)

Difficulty: Foundation / Marks: 3

Given $y = f(x)$ **with a maximum at** $(3, 5)$, **sketch** $y = f(2x)$.

Worked solution.

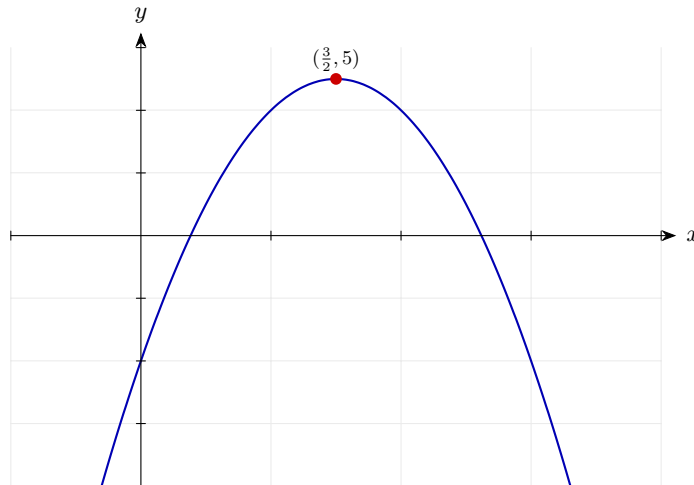
Horizontal compression by factor 2

$$(3, 5) \rightarrow \left(\frac{3}{2}, 5\right)$$

The transformation $y = f(2x)$ replaces x with $2x$, which compresses the graph horizontally by a factor of 2 (each feature happens at half the x -value). y -coordinates are unchanged, so the maximum at $(3, 5)$ becomes a maximum at $(\frac{3}{2}, 5)$.

Final answer: Same height, compressed horizontally; max at $(\frac{3}{2}, 5)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 64 (cg5-064)

Difficulty: Foundation / Marks: 3

Sketch $y = |2x - 4|$, showing the vertex and intercepts.

Worked solution.

Step 1: Vertex where $2x - 4 = 0$

$$x = 2; \quad \text{vertex } (2, 0)$$

The modulus function has its corner where the expression inside equals zero. Setting $2x - 4 = 0$ gives $x = 2$, and at that point $y = 0$.

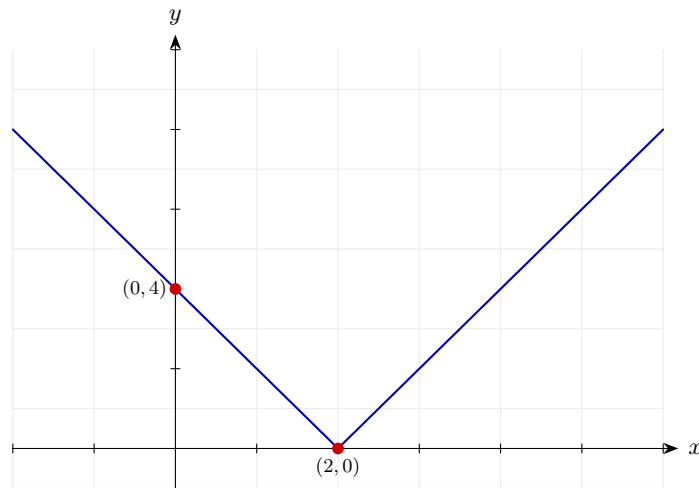
Step 2: y -intercept

$$y = |0 - 4| = 4$$

Substitute $x = 0$: $|-4| = 4$. Either side of the vertex the graph is a straight line with gradient ± 2 , giving the V-shape.

Final answer: V-shape; vertex $(2, 0)$; y -int $(0, 4)$; gradient ± 2

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 65 (cg5-065)

Difficulty: Foundation / Marks: 5

Sketch $y = -x^3 + 3x$, showing the stationary points and intercepts.

Worked solution.

Step 1: Intercepts

$$y = -x(x^2 - 3); \quad x = 0, \pm\sqrt{3}$$

Factor out $-x$, then $x^2 - 3 = 0$ gives $x = \pm\sqrt{3}$. Three distinct roots, with the curve passing through the origin.

Step 2: Stationary points

$$dy/dx = -3x^2 + 3 = 0 \implies x = \pm 1$$

Differentiate and set equal to zero. $3x^2 = 3 \Rightarrow x^2 = 1$, so $x = \pm 1$.

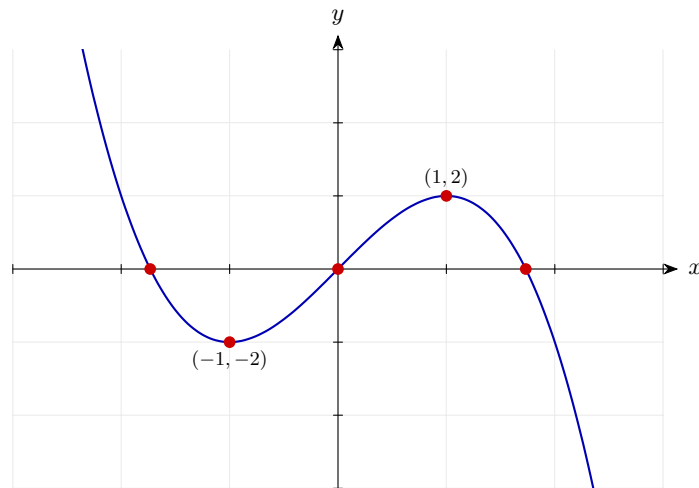
Step 3: Values

$$(-1, -2) \text{ min}; \quad (1, 2) \text{ max}$$

Substitute back: $y(1) = -1 + 3 = 2$ and $y(-1) = 1 - 3 = -2$. Negative leading coefficient (top-left to bottom-right), so the right-hand turning point is the maximum and the left-hand one is the minimum.

Final answer: Negative cubic; roots $0, \pm\sqrt{3}$; max $(1, 2)$; min $(-1, -2)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 66 (cg5-066)

Difficulty: Foundation / *Marks: 2*

Sketch $y = \ln x$, showing the x-intercept and asymptote.

Worked solution.

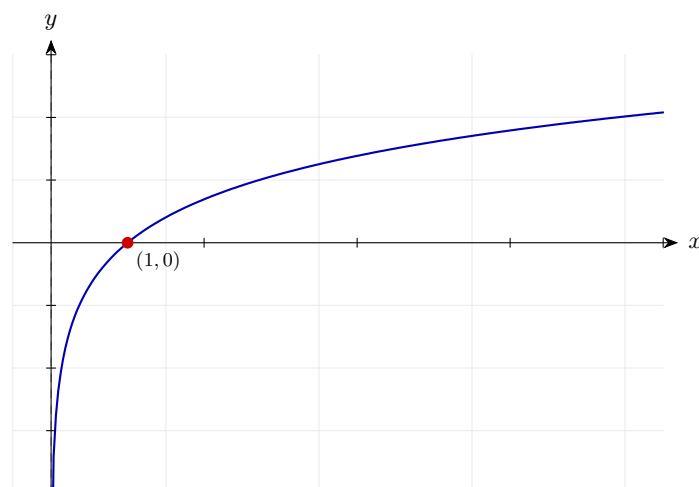
Key features

x -int: $(1, 0)$; asymptote $x = 0$; domain $x > 0$

$\ln x$ is defined only for $x > 0$, so the curve sits entirely to the right of the y -axis. $\ln 1 = 0$, giving the x -intercept at $(1, 0)$. As $x \rightarrow 0^+$, $\ln x \rightarrow -\infty$, giving the vertical asymptote $x = 0$. The curve increases (more slowly as x grows).

Final answer: Increasing curve; x -int $(1, 0)$; asymptote $x = 0$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 67 (cg5-067)

Difficulty: Foundation / Marks: 4

Sketch $y = (x - 1)^2(x + 2)$, showing all intercepts and where the curve touches the axis.

Worked solution.

Step 1: Roots

$$x = 1 \text{ (touches), } x = -2 \text{ (crosses)}$$

The factor $(x - 1)^2$ gives a double root at $x = 1$ (curve touches the axis), and $x + 2$ gives a single root at $x = -2$ (curve crosses).

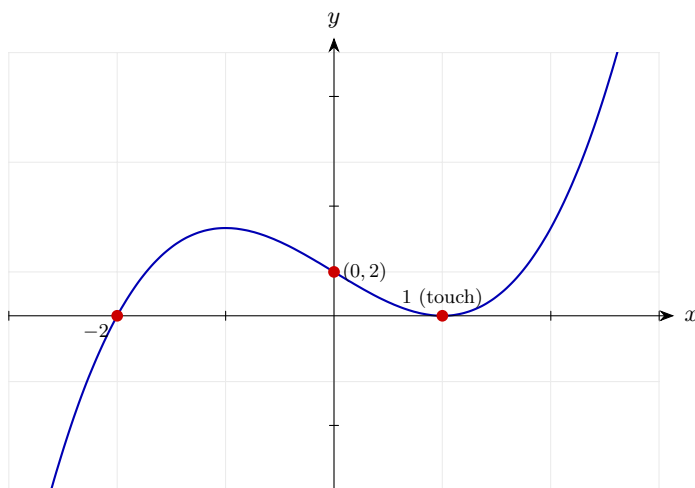
Step 2: y-intercept

$$y = (0 - 1)^2(0 + 2) = 1 \times 2 = 2$$

Substitute $x = 0$: $(-1)^2 = 1$ and $(0 + 2) = 2$. Positive leading coefficient gives a bottom-left to top-right cubic shape.

Final answer: Cubic; touches at $(1, 0)$; crosses at $(-2, 0)$; y-int $(0, 2)$

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 68 (cg5-068)

Difficulty: Foundation / Marks: 2

Sketch $y = \sin x$ for $0 \leq x \leq 2\pi$.

Worked solution.

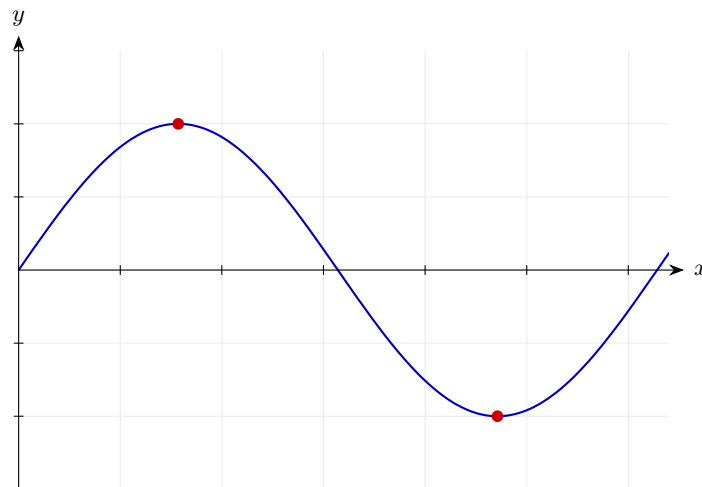
Key features

Starts at 0, max 1 at $\frac{\pi}{2}$, back to 0 at π , min -1 at $\frac{3\pi}{2}$

One full period of $\sin x$ has length 2π . It crosses the x -axis at $0, \pi, 2\pi$, peaks at $y = 1$ when $x = \frac{\pi}{2}$, and troughs at $y = -1$ when $x = \frac{3\pi}{2}$.

Final answer: Sine wave from 0 to 2π ; max 1, min -1

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 69 (cg5-069)

Difficulty: Foundation / Marks: 3

Given the graph of $y = f(x)$ passes through $(-1, 0)$, $(0, 2)$, $(3, 0)$, sketch $y = 2f(x)$.

Worked solution.

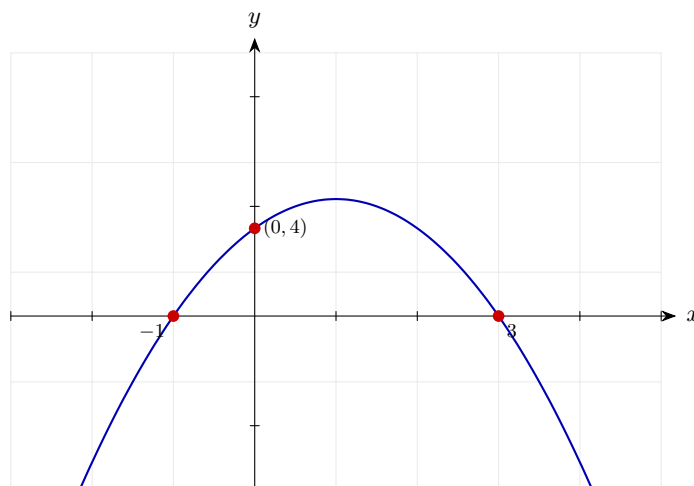
Vertical stretch by factor 2

$$(-1, 0) \rightarrow (-1, 0); \quad (0, 2) \rightarrow (0, 4); \quad (3, 0) \rightarrow (3, 0)$$

$y = 2f(x)$ multiplies every y -value by 2. Points on the x -axis stay fixed (since $2 \times 0 = 0$), so the roots are unchanged; the y -intercept doubles from 2 to 4.

Final answer: Same roots; y -int now $(0, 4)$; stretched vertically by 2

Sketch.



The curve plotted from its equation, with key features marked.

Curve Sketching 70 (cg5-070)

Difficulty: Foundation / Marks: 6

Sketch $y = x^3 - 6x^2 + 9x$, showing all intercepts, stationary points, and the point of inflection.

Worked solution.

Step 1: Factorise

$$y = x(x - 3)^2$$

Take out a common x to get $x(x^2 - 6x + 9)$, and recognise $x^2 - 6x + 9 = (x - 3)^2$ as a perfect square. The double factor means $x = 3$ is a double root.

Step 2: Stationary points

$$dy/dx = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0$$

Differentiate term by term and factor. Numbers multiplying to 3 and adding to -4 (after dividing by 3 inside the bracket) are -1 and -3 .

Step 3: Values

$$y(1) = 1 - 6 + 9 = 4; \quad y(3) = 0$$

Substitute back into the original equation. Positive leading coefficient: the left stationary point $(1, 4)$ is the maximum and the right one $(3, 0)$ is the minimum (the double root sits exactly on the x -axis).

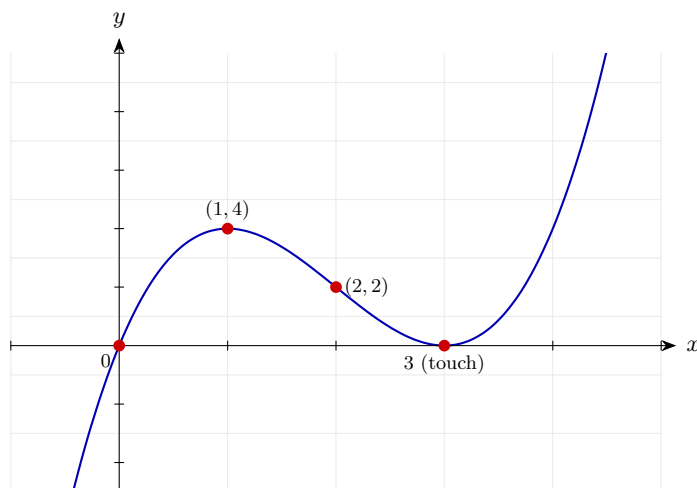
Step 4: Inflection

$$d^2y/dx^2 = 6x - 12 = 0 \implies x = 2; \quad y(2) = 2$$

A point of inflection occurs where the second derivative is zero and changes sign. $y(2) = 8 - 24 + 18 = 2$, so the inflection is at $(2, 2)$, midway between the two turning points.

Final answer: Cubic; roots 0, 3 (repeated); max $(1, 4)$; min $(3, 0)$; inflection $(2, 2)$

Sketch.



The curve plotted from its equation, with key features marked.
